Asset Price Dispersion, Monetary Policy and Macroprudential Regulation*

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This version: September 2026

Preliminary and incomplete draft

Abstract

Investors in financial markets often trade the same asset at different prices, and sellers sometimes receive no bid (a flash crash). One reason is that the price an investor trades at depends on how many brokers quote to that investor. This paper asks, "How do monetary policy and the regulation of intermediaries affect asset price dispersion and flash crash risk?" To answer this, it introduces price posting by brokers into a search-theoretic model of money and asset exchange. Higher inflation or tighter regulation reduces the number of bids, lowers equity prices and raises flash crash risk. The welfare-maximising level of flash crash risk is positive, because quotes are costly. Inflation shifts the gains from trade from sellers to buyers, so too few brokers bid, and the Friedman rule is optimal: without regulation, it brings flash crash risk down to its welfare-maximising level. Surprisingly, however, any regulation of intermediaries keeps flash crash risk above that level, even at the Friedman rule.

Keywords: Asset price dispersion; flash crashes; price posting; over-the-counter markets; search frictions; monetary policy; Friedman rule; macroprudential regulation.

JEL Classification: D43; D83; E31; E44; E52; G12; G24; G28.

*I am especially grateful to Sephorah Mangin and John Stachurski for their invaluable guidance, encouragement, and comments throughout the development of this paper. This paper was written as the third chapter of my PhD thesis at the Australian National University, and I gratefully acknowledge the generous financial support of the Australian National University, as well as the Australian Government for the research grants and stipends provided throughout my doctoral studies. All remaining errors are my own.

1 Introduction

On 6 May 2010, the prices of several U.S. equities collapsed to near zero and recovered within minutes, although nothing about their fundamentals had changed (Menkveld and Yueshen, 2019). In episodes like this, which are known as flash crashes, a seller who wants to trade finds no one willing to buy. They are rare, but not negligible: Jovanovic and Menkveld (2022) estimate that the probability that the order book for SPY, the exchange-traded fund that tracks the S&P 500, is empty averaged 0.018 percent, which implies a 40 percent chance of at least one empty book in their 11.5-year sample. More generally, investors often trade the same asset at different prices, and the distribution of the prices on offer has long tails.[1] These tails matter because the average price, or the average spread, says little about the risk faced by an investor who needs to sell: that risk depends on how often the market offers a poor price, or none at all.

The markets in which such risks arise have changed with policy since the 2007–2009 financial crisis. Basel III and the Dodd–Frank Act, and in particular its Volcker Rule, restricted the ability of banks to act as principals and raised the cost of making markets (Duffie, 2012, 2018). Jovanovic and Menkveld (2022) estimate that the number of middlemen bidding for SPY fell by more than half between 2007 and 2018, mostly around the implementation of the Volcker Rule in 2013–2014, and dealers increasingly avoid acting as principal (Bessembinder et al., 2018; Choi et al., 2024). Bid–ask spreads remained narrow, but trading large positions became more expensive, and bond markets became more prone to sudden breakdowns in intermediation capacity.[2] Over the same period, central banks expanded the money supply on an unprecedented scale (Gertler and Karadi, 2011; Chadha et al., 2020; Duc et al., 2022), and prolonged monetary accommodation has been linked to credit and asset price booms and, particularly when it is later reversed, to financial stress and crises (Grimm et al., 2023; Jiménez et al., 2026; Schularick et al., 2021; Boissay et al., 2023).

Search-theoretic models have become the standard way of studying over-the-counter trade and the role that money plays in it (Duffie et al., 2005; Lagos and Rocheteau, 2009; Lagos and Zhang, 2020). In most of these models, however, an investor who meets an intermediary

[1] See Zovko and Farmer (2002), Weber and Rosenow (2005) and Gu et al. (2008) on the distribution of limit-order prices and the shape of limit-order books, and Golub et al. (2012), Easley et al. (2011) and Gayduk and Nadtochiy (2018) on flash crashes.

[2] See Adrian et al. (2015, 2017), Anderson and Stulz (2017), Bao et al. (2018) and Dick-Nielsen and Rossi (2019), and Jovanovic and Menkveld (2024) on middlemen in limit order markets.

bargains over the terms of trade, so the price does not depend on how many intermediaries compete for the investor. There is no role for what this paper calls *quote intensity*, the mean number of quotes that an investor receives, and so no role for the dispersion of prices across investors, or for the event in which a seller receives no bid. Inventory models such as Grossman and Miller (1988) explain execution costs but not abrupt breakdowns in trading, and Jovanovic and Menkveld (2022) explain empty order books through costly participation, but for one side of the market and without money.

This paper introduces price posting by brokers into a search-theoretic model of money and asset exchange. Brokers enter freely to post asks and bids, investors receive a random number of quotes and trade at the best of them, as in Burdett and Judd (1983), and a cost of posting a quote stands in for the regulation of intermediaries. This paper asks three questions. How do inflation and regulation affect the dispersion of the prices at which investors trade? How do they affect the risk of a flash crash and the price of the asset? And which combination of monetary and regulatory policy is optimal?

The model answers these questions through a single variable. Free entry determines quote intensity on each side of the market in closed form: brokers post quotes until the expected spread of the marginal quote equals its cost. Inflation lowers the real value of money, and with it the real spread that a bid earns, while regulation raises the cost of posting a quote. Quote intensity is therefore the channel through which both policies reach the distribution of executable prices, the probability of a flash crash and welfare, which allows the two policies to be compared directly.

Main results. First, the option to resell makes equity worth more than its fundamental value. Higher inflation and tighter regulation lower the real value of money and the real price that a seller can expect, and with them the equity price and this speculative premium (Propositions 3, 4 and 5). The premium is always positive: because the bottom of the book is the seller's reservation price, the option to resell can never be worth less than nothing.

Second, inflation and regulation both reduce the number of bids, but they differ on the ask side of the market. One might expect inflation to thin the market on both sides. Instead, it draws in more asks even as it drives out bids, because it shifts the gain from trade at the interbroker price from sellers to buyers, whereas regulation reduces quoting on both sides (Proposition 6). In both cases, executable asks shift up and real executable bids shift down (Proposition 7).

Third, flash crashes arise endogenously, without any shock to fundamentals, and their probability governs the tail of executable bids: at any confidence level below the probability of a flash crash, the Value-at-Risk of a seller is the bottom of the book (Proposition 9 and Corollary 3). Higher inflation and tighter regulation both make flash crashes more likely, but they reshape the bid side of the book differently. Tighter regulation spreads executable bids towards the bottom of the book, whereas inflation also raises sellers' reservation price towards the interbroker price, so that a flash crash, although more likely, leaves a seller's price closer to the interbroker price.

Fourth, the welfare-maximising probability of a flash crash is positive, because quotes are costly (Proposition 10). Inflation shifts the gains from trade from sellers to buyers, so too few brokers bid, and it draws in asks that use resources without improving the allocation. The Friedman rule is therefore optimal: without regulation, it brings bid quoting, and with it flash crash risk, to their welfare-maximising levels. Surprisingly, however, any regulation of intermediaries keeps flash crash risk above its welfare-maximising level, even at the Friedman rule, and tighter regulation pushes it further above.

Other results. An extension after the references replaces the Poisson distribution of quotes with the negative binomial, which allows quotes to be overdispersed, as they are in the data. Overdispersion can raise the mean number of quotes, yet it makes flash crashes more likely, and reducing it, for example through market design, lowers flash crash risk even at the Friedman rule. The extension also characterises the welfare-maximising degree of overdispersion when market design is costly.

Related literature. The closest paper is Jovanovic and Menkveld (2022), who adapt the auction of Levin and Smith (1994) to explain the shape of order-book bids and the possibility that no bid arrives, building on common-value and all-pay auctions (Hausch and Li, 1993; Baye et al., 1996). The bid distribution in this paper has the same form as theirs, and their estimated number of bidders is the empirical counterpart of bid intensity (Section 7). Their model, however, has one side of the market and no money. Relative to Jovanovic and Menkveld (2022), this paper models both sides of the market and links quote intensity to monetary policy, asset prices and welfare. Earlier work derives price dispersion from uncertain demand or captive customers (Prescott, 1975; Shilony, 1977; Rosenthal, 1980; Varian, 1980).

Price dispersion in the model comes from noisy search and costly posting, as in Burdett and Judd (1983), and this paper uses the tractable Poisson form of Mortensen (2005) and Wang

et al. (2020). Head et al. (2012) and Burdett et al. (2015) use this framework to generate dispersed and sticky prices in monetary economies, and Bajaj and Mangin (2026) show that when buyers can meet several sellers and choose among them, greater choice can raise or lower market power and can make inflation more costly. In all of these papers, sellers post terms of trade for goods. Here, brokers post prices on both sides of an asset market, and the value of money determines how the gain from trade is split between them.

This paper also builds on monetary models of over-the-counter (OTC) trade, in particular Lagos and Zhang (2020), which adapts the OTC market of Duffie et al. (2005) to the monetary framework of Lagos and Wright (2005). Related models include Lagos and Zhang (2019a,b), Mattesini and Nosal (2016) and Geromichalos and Jung (2019).[3] In these models, assets earn resale premia when investors expect to sell to others who value them more (Lagos and Zhang, 2019a,b, 2020). With price posting in place of bargaining, this paper shows how the resale premium depends on how actively brokers bid, and studies how monetary policy affects the efficiency of intermediation, a question that Geromichalos and Jung (2019) also address.

Finally, this paper relates to work on how post-crisis regulation affects market making, including Duffie (2012, 2018) and Trebbi and Xiao (2019). Relative to these papers, which study regulation on its own, this paper studies how it interacts with monetary policy, and how the two policies together determine the probability of a flash crash.

Outline. Section 2 describes the model, and Section 3 characterises equilibrium. Section 4 studies asset prices and the speculative premium, and Section 5 studies price dispersion and flash crashes. Section 6 examines monetary and macroprudential policy. Section 7 relates the results to the empirical evidence, and Section 8 concludes. The extension on overdispersed quote arrivals follows the references. Proofs and supplementary derivations are available from the author on request.

2 Model

Time is represented by a sequence of periods indexed by $t = 0, 1, \ldots$ Each time period is divided into two subperiods in which different activities take place. There is a continuum of

[3] See Lagos et al. (2017) for a survey. Assets can also earn liquidity premia as media of exchange or through their resale for money (Geromichalos et al., 2007; Lagos and Rocheteau, 2009; Lagos, 2010; Jacquet and Tan, 2012; Lester et al., 2012; Nosal and Rocheteau, 2013; Geromichalos et al., 2021, 2023), money can constrain the market power of intermediaries (Lagos and Zhang, 2022), and liquidity and asset-price volatility can be determined jointly (Rocheteau and Wang, 2023).

infinitely lived agents called *investors*, each identified with a point in the set $\mathcal{I} = [0, 1]$. There is also a continuum of infinitely lived agents called *asking brokers*, each identified with a point in the set $\mathcal{B}^A = [0, 1]$, and a continuum of infinitely lived agents called *bidding brokers*, each identified with a point in the set $\mathcal{B}^B = [0, 1]$.[4] All agents discount across periods with factor $\beta \in (0, 1)$.

There is a continuum of production units with measure $A^s \in \mathbb{R}_{++}$ that are active every period. Every active unit yields an exogenous *dividend* $y_t \in \mathbb{R}_+$. Each active unit yields the same dividend, therefore $y_t A^s$ is the aggregate dividend. At the beginning of every period, each active unit is subject to an independent idiosyncratic shock that renders it permanently unproductive with probability $1 - \eta \in [0, 1)$. If a production unit remains active, its dividend in period t is $y_t = \gamma_t y_{t-1}$, where γ_t is a nonnegative random variable with cumulative distribution function Φ, i.e., $\mathbb{P}(\gamma_t \leq \gamma) = \Phi(\gamma)$, and mean $\bar{\gamma} \in (0, (\beta\eta)^{-1})$.

The time-t dividend becomes public knowledge at the beginning of period t. At that time each failed production unit is replaced by a new unit that pays y_t in the initial period and follows the same process as other active units thereafter. The dividend of the initial set of production units, $y_0 \in \mathbb{R}_{++}$, is given at $t = 0$. In the second subperiod of every period, every agent has access to a linear production technology that transforms effort into a perishable homogeneous consumption good.

For each active production unit there exists a durable and perfectly divisible *equity share* that represents ownership of the production unit and entitles the holder to its dividends. At the start of every period $t \geq 1$, each investor receives an endowment of $(1 - \eta)A^s$ equity shares associated with the new production units (when a production unit fails, its equity share is extinguished). There is a second financial instrument, *money*, that is intrinsically useless. The quantity of money at time t is denoted A_t^m. The initial quantity of money, $A_0^m \in \mathbb{R}_{++}$, is given, and $A_{t+1}^m = \mu A_t^m$, with $\mu \in \mathbb{R}_{++}$. A monetary authority injects or withdraws money via lump-sum transfers or taxes to investors in the second subperiod of each period. At the beginning of period $t = 0$, each investor is endowed with a portfolio of equity shares and money. All financial instruments are perfectly recognisable, not forgeable, and can be traded among agents in every subperiod.

[4] It is possible to model asking and bidding brokers as a single representative broker type that performs both functions. However, such an approach complicates the exposition and obscures the distinction between the two sides of the market. For clarity, this paper treats asking and bidding brokers as separate agent types. This separation is purely expositional and does not affect the equilibrium results.

In the second subperiod of every period, all investors can trade the consumption good produced in that subperiod, equity shares, and money in a spot Walrasian market. In the first subperiod of every period, trading is organised as follows: brokers post nominal price quotes p, taking as given the distribution of prices posted by all other brokers (for asking brokers, this is described by the cumulative distribution function (CDF) $A_t(p)$, and for bidding brokers, by the CDF $B_t(p)$). Investors know the distributions $A_t(p)$ and $B_t(p)$ but can only contact, and hence can only trade with, a random sample of brokers. The number of quotes that an investor observes is a discrete random variable $k \in \{0, 1, 2, \dots\}$. A search technology $P_k(\theta)$ gives the probability that an investor observes k quotes when the mean number of quotes is $\theta > 0$, so that $\sum_{k\geq 0} P_k(\theta) = 1$ and $\sum_{k\geq 0} kP_k(\theta) = \theta$. The mean can differ between the two sides of the market: this paper writes θ_a for the mean number of asks and θ_b for the mean number of bids that an investor observes. Once an investor and broker have met, the transaction takes place at the price posted by the broker. Brokers act as intermediaries, executing trades on behalf of investors in the decentralised market and offsetting them with other brokers in an interbroker market. After the transaction is completed, the investor and broker part ways.[5] The timing assumption is that the round of trade between investors and brokers takes place in the first subperiod of a typical period t, and ends before the equity shares yield dividends.[6] Hence, equity is traded *cum-dividend* in the first subperiod, but *ex-dividend* in the Walrasian market of the second subperiod. This paper assumes that agents cannot make binding commitments, that there is no enforcement, and that histories of actions are private, which precludes any borrowing or lending, so all trade must be *quid pro quo*. This assumption, together with the structure of preferences described below, creates the need for a medium of exchange.[7,8]

A broker's preferences are given by

$$\mathbb{E}_0^B \sum_{t=0}^{\infty} \beta^t (c_t - h_t),$$

[5] See Zhang (2018) for an OTC model with long-term relationships between investors and dealers.

[6] As in previous search models of OTC markets, e.g., Duffie et al. (2005) and Lagos and Rocheteau (2009), an investor must own the equity in order to consume the dividend flow of the consumption good in the first subperiod.

[7] Under these conditions, there cannot exist a futures market for the equity share or the dividend, so an agent who wishes to consume the dividend must hold the equity share at the time the dividend is paid. A similar assumption is typically made in search models of financial OTC trade, e.g., Duffie et al. (2005), Lagos and Rocheteau (2009), Lagos and Zhang (2019b), or Lagos and Zhang (2020).

[8] See Lagos and Zhang (2019b) for a similar model where investors can buy equity with *margin loans*.

where c_t and h_t denote a broker's consumption of the homogeneous good that is produced, traded, and consumed in the second subperiod of period t, and the utility cost of exerting h_t units of effort to produce this good.[9] The expectation operator $\mathbb{E}_0^B$ is with respect to the probability measure induced by the random trading process in the first subperiod.

An investor's preferences are given by

$$\mathbb{E}_0^I \sum_{t=0}^{\infty} \beta^t (\varepsilon_i y_t + c_t - h_t),$$

where c_t is consumption of the homogeneous good that is produced, traded, and consumed in the second subperiod of period t, and h_t is the utility cost of exerting h_t units of effort to produce this good. The variable y_t is the quantity of the dividend good that an investor consumes at the end of the first subperiod of period t, and ε_i denotes the realisation of a preference shock that is distributed independently over time and across agents, with cumulative distribution function G, where $\varepsilon_i \in \{\varepsilon_L, \varepsilon_H\}$ with corresponding probabilities $\pi_i \in (0,1)$ and $\bar{\varepsilon} = \sum_{i \in \{L,H\}} \pi_i \varepsilon_i$. An investor learns their realisation ε_i at the beginning of period t, before trading has commenced. The expectation operator $\mathbb{E}_0^I$ is with respect to the probability measure induced by the dividend process, the investor's preference shock, and the random trading process in the first subperiod of each period t.

3 Equilibrium

3.1 Investors

Consider the determination of allocations in the first subperiod of period t for an investor who enters the period with portfolio $\mathbf{a}_t \equiv (a_t^m, a_t^s)$, consisting of $a_t^m \in \mathbb{R}_+$ units of money and $a_t^s \in \mathbb{R}_+$ units of equity, and preference type ε_i.

Let $W_t(\mathbf{a}_t)$ denote the maximum expected discounted payoff of an investor holding portfolio $\mathbf{a}_t$ at the beginning of the second subperiod of period t, after dividends have been realised. Suppose the investor enters period t with (pre-trade) portfolio $\mathbf{a}_t \equiv (a_t^m, a_t^s)$ and preference type ε_i. If the investor meets a broker, then they are able to trade and their post-trade portfolio is

[9] Brokers receive no utility from dividends and hold no equity, so they have no portfolio problem and no motive to trade equity on their own account. This assumption is standard in search models of OTC markets, e.g., Duffie et al. (2005) or Lagos and Rocheteau (2009); see Weill (2007) or Lagos and Zhang (2015) for models in which dealers hold assets. Because brokers do not trade on their own account, this paper proxies trading activity by quote intensity rather than by trade volume.

$\bar{\mathbf{a}}_t(\mathbf{a}_t, \varepsilon_i) \equiv (\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i), \bar{a}_t^s(\mathbf{a}_t, \varepsilon_i))$, and,

$$(\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i), \bar{a}_t^s(\mathbf{a}_t, \varepsilon_i)) = \underset{(\bar{a}_t^m, \bar{a}_t^s) \in \mathbb{R}_+^2}{\arg\max} \ \varepsilon_i y_t \bar{a}_t^s + W_t(\bar{\mathbf{a}}_t) \tag{1}$$
$$\text{s.t. } \bar{a}_t^m + p\bar{a}_t^s \leq a_t^m + pa_t^s.$$

where p is the dollar price of equity in the first subperiod of period t.

This paper now analyses the investor's problem in a typical period. Let $V_t(\mathbf{a}_t, \varepsilon_i)$ denote the maximum expected discounted payoff of an investor with type ε_i and portfolio $\mathbf{a}_t = (a_t^m, a_t^s)$ at the beginning of the first subperiod of period t. Let ϕ_t^m be the real price of money, and ϕ_t^s be the *ex-dividend* price of equity in the second subperiod of period t (both expressed in terms of the second subperiod consumption good). In the second subperiod, the investor chooses consumption of the homogeneous good c_t, the utility cost of production h_t, and the end-of-subperiod portfolio $\tilde{\mathbf{a}}_{t+1} \equiv (\tilde{a}_{t+1}^m, \tilde{a}_{t+1}^s)$, recognising that at the end of the second subperiod a fraction $1 - \eta$ of equity is rendered unproductive and replaced. Therefore,

$$W_t(\mathbf{a}_t) = \max_{(c_t, h_t, \tilde{\mathbf{a}}_{t+1}) \in \mathbb{R}_+^4} \left[c_t - h_t + \beta \mathbb{E}_t \sum_{i \in \{L,H\}} \pi_i V_{t+1}(\mathbf{a}_{t+1}, \varepsilon_i) \right] \tag{2}$$
$$\text{s.t. } c_t + \boldsymbol{\phi}_t \tilde{\mathbf{a}}_{t+1} \leq h_t + \boldsymbol{\phi}_t \mathbf{a}_t + T_t$$
$$\mathbf{a}_{t+1} = \left[\tilde{a}_{t+1}^m, \eta \tilde{a}_{t+1}^s + (1 - \eta) A^s \right],$$

where $\mathbb{E}_t$ is the conditional expectation over the next-period realisation of the dividend, T_t is the real value of the time-t lump-sum monetary transfer (a tax if negative), $\boldsymbol{\phi}_t \equiv (\phi_t^m, \phi_t^s)$, and $\boldsymbol{\phi}_t \mathbf{a}_t$ denotes the dot product of $\boldsymbol{\phi}_t$ and $\mathbf{a}_t$. Since ε_i is independently and identically distributed over time, $W_t(\mathbf{a}_t)$ is independent of ε_i, and the portfolio that each investor chooses to carry into period $t + 1$ is independent of ε_i.

The value function of an investor who enters the first subperiod of period t with portfolio $\mathbf{a}_t$ and preference type ε_i can now be written. An investor who observes no quote consumes the dividend on the equity they hold and enters the second subperiod with an unchanged portfolio, and an investor who observes $k \geq 1$ quotes trades at the best of them. A high-valuation investor, who buys equity (Lemma 1 below), trades at the lowest of k asks, which

has distribution $1-[1-A_t(p)]^k$, where A_t is the distribution of ask prices; a low-valuation investor, who sells, trades at the highest of k bids, which has distribution $[B_t(p)]^k$, where B_t is the distribution of bid prices. Hence

$$V_t(\mathbf{a}_t,\varepsilon_i)=\begin{cases} P_0(\theta_a)\left[\varepsilon_H y_t a_t^s+W_t(\mathbf{a}_t)\right] & \\ \quad+\sum_{k=1}^{\infty}kP_k(\theta_a)\int\left\{\varepsilon_H y_t\bar{a}_t^s(\mathbf{a}_t,\varepsilon_H)+W_t\left[\bar{\mathbf{a}}_t(\mathbf{a}_t,\varepsilon_H)\right]\right\} & \\ \quad\times[1-A_t(p)]^{k-1}\,\mathrm{d}A_t(p) & \text{if } \varepsilon_i=\varepsilon_H, \\ P_0(\theta_b)\left[\varepsilon_L y_t a_t^s+W_t(\mathbf{a}_t)\right] & \\ \quad+\sum_{k=1}^{\infty}kP_k(\theta_b)\int\left\{\varepsilon_L y_t\bar{a}_t^s(\mathbf{a}_t,\varepsilon_L)+W_t\left[\bar{\mathbf{a}}_t(\mathbf{a}_t,\varepsilon_L)\right]\right\} & \\ \quad\times[B_t(p)]^{k-1}\,\mathrm{d}B_t(p) & \text{if } \varepsilon_i=\varepsilon_L, \end{cases} \tag{3}$$

where $k[1-A_t(p)]^{k-1}\mathrm{d}A_t(p)$ and $k[B_t(p)]^{k-1}\mathrm{d}B_t(p)$ are the densities of the lowest of k asks and the highest of k bids.

3.2 Brokers

Next, this paper analyses the brokers' problem. Posting a quote costs a broker κ_t units of the consumption good for each share that the quote covers, paid in the second subperiod, and entry into quoting is free: any broker can post any number of quotes at this cost.[10] A quote covers the whole trade that its recipient wants to make: by Lemma 1 below, an ask at price p covers the a_t^m/p shares that a buyer's money buys, and a bid covers the a_t^s shares that a seller holds. Ask quotes are observed by high-valuation investors and bid quotes by low-valuation investors, as in (3). Brokers offset the trades they execute at the interbroker price $\hat{p}$: an asking broker who sells a share to an investor at p buys it from a bidding broker at $\hat{p}$, and a bidding broker who buys a share at p sells it at $\hat{p}$.[11] An executed ask at price p therefore earns the nominal spread $p-\hat{p}$ on each share, worth $\phi_t^m(p-\hat{p})$ units of the consumption good, and an executed bid at price p earns $\phi_t^m(\hat{p}-p)$ on each share. Because the spread and the posting cost

[10] The posting cost captures the operating, capital and compliance costs of making a market, which scale with the size of the position that a quote commits the broker to. Section 6 separates the resource cost of quoting a share from a regulatory wedge.

[11] Brokers therefore hold no equity, and the interbroker market clears whenever the first-subperiod market for equity does. They receive their spreads in money and spend them on the consumption good in the second subperiod. The numerical example below chooses the units of nominal prices so that $\hat{p}=1$.

are both per share, the number of shares that a quote covers scales its revenue and its cost alike and drops out of the brokers' problem. Investors accept a quote only if it is at least as good as keeping their portfolio. By Lemma 1 below, a high-valuation investor buys at an ask p only if $p \leq \bar{p}_t \equiv (\varepsilon_H y_t + \phi_t^s)/\phi_t^m$, the price at which the cutoff ε_t^* equals ε_H, and a low-valuation investor sells at a bid p only if $p \geq \underline{p}_t \equiv (\varepsilon_L y_t + \phi_t^s)/\phi_t^m$, the price at which $\varepsilon_t^* = \varepsilon_L$. These reservation prices are the edges of the book: an ask above $\bar{p}_t$ or a bid below $\underline{p}_t$ never executes.

An ask quote executes if it is the lowest ask observed by the investor who receives it. A given ask reaches an investor who observes k asks in total with probability $kP_k(\theta_a)/\theta_a$, where θ_a is the mean number of asks that a high-valuation investor observes, and it is the lowest of them with probability $[1 - A_t(p)]^{k-1}$. Because brokers' preferences are linear and they carry no assets from one period to the next, a quote posted in period t does not affect a broker's payoff in any other period, so the value of posting it is its expected profit. Free entry drives this profit to zero. Per share covered, for asking brokers,

$$\Pi_{a,t}(p) \equiv \sum_{k=1}^{\infty} \frac{kP_k(\theta_a)}{\theta_a} [1 - A_t(p)]^{k-1} \phi_t^m (p - \hat{p}) - \kappa_t \leq 0 \quad \text{for all } p \in [\hat{p}, \bar{p}_t], \tag{4}$$

with equality on the support of A_t, denoted $\mathcal{A}_t$, and, by the same argument, for bidding brokers,

$$\Pi_{b,t}(p) \equiv \sum_{k=1}^{\infty} \frac{kP_k(\theta_b)}{\theta_b} [B_t(p)]^{k-1} \phi_t^m (\hat{p} - p) - \kappa_t \leq 0 \quad \text{for all } p \in [\underline{p}_t, \hat{p}], \tag{5}$$

with equality on the support of B_t, denoted $\mathcal{B}_t$, where θ_b is the mean number of bids that a low-valuation investor observes. A higher ask earns a larger spread if it executes but is less likely to be the lowest ask the investor observes, and a lower bid trades off in the same way. Free entry requires every price that brokers post to earn, in expectation, exactly the cost of posting it, and no price in the book to earn more. Brokers therefore earn zero expected profit, and the mean numbers of quotes, θ_a and θ_b, and with them the probabilities $P_k(\theta_a)$ and $P_k(\theta_b)$, are determined in equilibrium.

3.3 Definition of Equilibrium

Let A_{It}^m and A_{It}^s denote the quantities of money and equity shares, respectively, held by all investors at the beginning of the period t (after production units have depreciated and been

replaced). That is, $A_{It}^m = \int a_t^m \mathrm{d}H_t(\mathbf{a}_t)$ and $A_{It}^s = \int a_t^s \mathrm{d}H_t(\mathbf{a}_t)$, where H_t is the cumulative distribution function over portfolios $\mathbf{a}_t = (a_t^m, a_t^s)$ held by investors at the beginning of the first subperiod of period t. Let $\tilde{A}_{It+1}^m$ and $\tilde{A}_{It+1}^s$ denote the total quantities of money and shares held by all investors at the end of period t, i.e., $\tilde{A}_{It+1}^h = \int_{\mathcal{I}} \tilde{a}_{it+1}^h \mathrm{d}i$ for $h \in \{s, m\}$ with $A_{It+1}^m = \tilde{A}_{It+1}^m$, and $A_{It+1}^s = \eta \tilde{A}_{It+1}^s + (1-\eta)A^s$. Let $\bar{A}_{It}^m$ and $\bar{A}_{It}^s$ denote the quantities of money and shares held by all investors after the first subperiod of period t, including the investors who receive no quote and keep their portfolios, and let $\bar{A}_{Bt}^m$ denote the money that brokers hold at that point, which consists of the spreads they have received. For asset $h \in \{s, m\}$,

$$\begin{aligned}
\bar{A}_{It}^h = & \left[\pi_H P_0(\theta_a) + \pi_L P_0(\theta_b)\right] A_{It}^h \\
& + \pi_H \int \sum_{k=1}^{\infty} k P_k(\theta_a) \int \bar{a}_t^h(\mathbf{a}_t, \varepsilon_H) \left[1 - A_t(p)\right]^{k-1} \mathrm{d}A_t(p) \mathrm{d}H_t(\mathbf{a}_t) \\
& + \pi_L \int \sum_{k=1}^{\infty} k P_k(\theta_b) \int \bar{a}_t^h(\mathbf{a}_t, \varepsilon_L) \left[B_t(p)\right]^{k-1} \mathrm{d}B_t(p) \mathrm{d}H_t(\mathbf{a}_t).
\end{aligned}$$

Equilibrium can now be defined.

Definition 1. *An* equilibrium *consists of sequences of prices* $\{\phi_t^m, \phi_t^s\}_{t=0}^{\infty}$*, of ask and bid distributions* $\{A_t(p), B_t(p)\}_{t=0}^{\infty}$*, mean numbers of asks and bids* $\{\theta_a, \theta_b\}$*, first-subperiod portfolios* $\{\bar{\mathbf{a}}_t(\cdot)\}_{t=0}^{\infty}$ *and end-of-period portfolios* $\{\tilde{\mathbf{a}}_{t+1}\}_{t=0}^{\infty}$ *such that, for all* t*:*

(i) the first-subperiod portfolios solve (1)*;*

(ii) taking prices and first-subperiod portfolios as given, the end-of-period portfolio solves (2)*;*

(iii) all markets clear: the end-of-period Walrasian market for equity, $\tilde{A}_{It+1}^s = A^s$*, the first-subperiod market for equity,* $\bar{A}_{It}^s = A^s$*, and the first-subperiod market for money,* $(\bar{A}_{It}^m + \bar{A}_{Bt}^m - A_t^m)\mathbb{I}_{\{\phi_t^m > 0\}} = 0$*, where* $\mathbb{I}_{\{\cdot\}}$ *is an indicator function; and*

(iv) taking prices and distributions as given, quotes satisfy the free-entry conditions (4) *and* (5)*.*

The first step toward characterising equilibrium is to find the post-trade portfolio allocations of the investor. The maximisation problem in (1) represents the portfolio problem of an investor with price p in the first subperiod of period t. The solution is summarised as follows:

Lemma 1. *Define* $\varepsilon_t^* \equiv \frac{p\phi_t^m - \phi_t^s}{y_t}$ *and*

$$\chi(\varepsilon_t^*, \varepsilon_i) \begin{cases} = 1 & \text{if } \varepsilon_t^* < \varepsilon_i \\ \in [0,1] & \text{if } \varepsilon_t^* = \varepsilon_i \\ = 0 & \text{if } \varepsilon_t^* > \varepsilon_i. \end{cases} \tag{6}$$

Consider an investor who enters the first subperiod of period t *with portfolio* $\mathbf{a}_t$ *and valuation* ε_i*. The investor's post-trade portfolio,* $[\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i), \bar{a}_t^s(\mathbf{a}_t, \varepsilon_i)]$*, is given by*

$$\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i) = [1 - \chi(\varepsilon_t^*, \varepsilon_i)]\,(a_t^m + pa_t^s) \tag{7}$$

$$\bar{a}_t^s(\mathbf{a}_t, \varepsilon_i) = \chi(\varepsilon_t^*, \varepsilon_i)(1/p)(a_t^m + pa_t^s). \tag{8}$$

Lemma 1 provides a complete characterisation of the post-trade portfolios of investors in the first subperiod of period t. The outcome depends on whether the investor's valuation ε_i is above or below a cutoff level ε_t^*. That is, investors with preference type $\varepsilon_i > \varepsilon_t^*$ use all their cash to purchase equity, whereas if $\varepsilon_i < \varepsilon_t^*$, they sell all of their equity holdings for cash.[12]

3.4 Recursive Equilibrium and Price Dispersion

In what follows, this paper specialises the analysis to recursive monetary equilibria. That is, equilibria in which asset holdings are constant over time, i.e., $A_{It}^s = A_I^s$, nominal prices grow with the money stock,[13] and real asset prices are time-invariant functions of the aggregate dividend, i.e., $\phi_t^s = \phi^s y_t$, $p\nu_t\phi_t^m \equiv \bar{\phi}_t^s = \bar{\phi}^s y_t = p\varphi^m y_t$, $\kappa_t = \kappa y_t$, $\phi_t^m A_{It}^m = ZA^s y_t$, where $Z \in \mathbb{R}_+$. Hence, in a recursive monetary equilibrium $\varepsilon_t^* = p\varphi^m - \phi^s \equiv \varepsilon^*$, $\phi_{t+1}^s/\phi_t^s = \gamma_{t+1}$, $\phi_t^m/\phi_{t+1}^m = \mu/\gamma_{t+1}$, and the reservation prices are $\bar{p} = (\varepsilon_H + \phi^s)/\varphi^m$ and $\underline{p} = (\varepsilon_L + \phi^s)/\varphi^m$. Throughout the analysis, this paper lets $\bar{\beta} \equiv \beta\bar{\gamma}$, $\bar{\beta}_L \equiv \pi_L\bar{\beta}$, and $\bar{\beta}_H \equiv \pi_H\bar{\beta}$ and maintains the following assumption,

Assumption 1. *The number of asks* $k \in \{0, 1, 2, \dots\}$ *received by a high-valuation investor*

[12] Given ε_t^*, $p\phi_t^m = \varepsilon_t^* y_t + \phi_t^s$ can be interpreted as the *cum-dividend* real value of equity to the marginal investor in period t.

[13] Because the money stock grows at rate μ, nominal prices must grow at the same rate in a recursive equilibrium. This paper therefore measures posted prices, the interbroker price and the book in units of ν_t dollars, where $\nu_{t+1} = \mu\nu_t$: a quote p is a price of $p\nu_t$ dollars in period t, so the interbroker price $\hat{p}$ and the book $[\underline{p}, \bar{p}]$ are fixed in these units even though the money stock grows, and $\varphi^m y_t \equiv \nu_t\phi_t^m$ is the real value of one unit. All nominal magnitudes below, including the widths of the book, nominal spreads and the quantiles of executable prices, are measured in these units. The money stock measured in the same units, $A_t^m/\nu_t = ZA^s/\varphi^m$, is constant along a recursive equilibrium.

and the number of bids received by a low-valuation investor are Poisson distributed with means $\theta_a \in \mathbb{R}_+$ *and* $\theta_b \in \mathbb{R}_+$*. That is, the probability of receiving* k *quotes on side* $j \in \{a, b\}$ *is given by*

$$P_k(\theta_j) = \frac{\theta_j^k e^{-\theta_j}}{k!}. \tag{9}$$

Following Wang et al. (2020) and Mortensen (2005), this paper assumes that the number of quotes observed by each investor follows a Poisson distribution. This is the distribution that arises when a large number of quotes is allocated at random across a large number of investors, and θ_a and θ_b are then the numbers of asks and bids posted per investor on each side of the market, which free entry determines. This paper calls them the *quote intensities* of the ask and bid sides. With Poisson sampling, the number of other quotes observed by the investor who receives a given quote is also Poisson with the same mean, so an ask quote at price p executes with probability $e^{-\theta_a A(p)}$ and a bid quote with probability $e^{-\theta_b[1-B(p)]}$. The following result characterises the equilibrium ask and bid price distributions, denoted by $A(p)$ and $B(p)$, respectively, together with the quote intensities implied by free entry. Building on the framework of Burdett and Judd (1983), these distributions describe the pricing behaviour of brokers in a decentralised market with random search and costly quotes.

Proposition 1. *If* $P_k(\theta)$ *is Poisson with mean* $\theta \in \mathbb{R}_+$*, then*

(i) There exists a unique ask price distribution consistent with free entry of asking brokers, given by

$$A(p) = \frac{1}{\theta_a} \ln \left[\frac{\varphi^m (p - \hat{p})}{\kappa} \right], \tag{10}$$

with support $\mathcal{A} \equiv [\hat{p} + \kappa/\varphi^m, \bar{p}] \subset \mathbb{R}_+$ *and the corresponding density is*

$$a(p) = \frac{1}{\theta_a (p - \hat{p})}. \tag{11}$$

(ii) There exists a unique bid price distribution consistent with free entry of bidding brokers,

given by

$$B(p) = 1 - \frac{1}{\theta_b} \ln \left[\frac{\varphi^m (\hat{p} - p)}{\kappa} \right], \tag{12}$$

with support $\mathcal{B} \equiv \left[\underline{p}, \hat{p} - \kappa/\varphi^m\right] \subset \mathbb{R}_+$ *and the corresponding density is*

$$b(p) = \frac{1}{\theta_b (\hat{p} - p)}. \tag{13}$$

(iii) Free entry determines the equilibrium quote intensities,

$$\theta_a = \ln \left(\frac{\varphi^m \Gamma_a}{\kappa} \right) \quad \textit{and} \quad \theta_b = \ln \left(\frac{\varphi^m \Gamma_b}{\kappa} \right), \tag{14}$$

where $\Gamma_a \equiv \bar{p} - \hat{p}$ *and* $\Gamma_b \equiv \hat{p} - \underline{p}$ *are the widths of the ask and bid sides of the book.*

An asking broker faces a trade-off between posting a higher price, which earns a larger spread if the quote executes, and posting a lower price, which is more likely to execute. The same logic applies symmetrically to bidding brokers. The equilibrium distributions $A(p)$ and $B(p)$ make every price in their supports earn the same expected spread, and free entry sets that expected spread equal to the posting cost. Price dispersion therefore arises endogenously from brokers' indifference across the prices they post.[14]

Free entry pins down both the quote intensities and the supports. A quote at the top of the book, $\bar{p}$, executes only if the investor observes no other quote, which happens with probability $e^{-\theta_a}$, so free entry requires $e^{-\theta_a}\varphi^m\Gamma_a = \kappa$; similarly, a quote at the bottom of the book, $\underline{p}$, requires $e^{-\theta_b}\varphi^m\Gamma_b = \kappa$, which is part *(iii)*. A quote at the bottom of the support of $A(p)$ executes for sure, so it breaks even only if the real spread just covers the posting cost, which gives the lowest ask $\hat{p} + \kappa/\varphi^m$; symmetrically, the highest bid is $\hat{p} - \kappa/\varphi^m$. Given the equity price, a higher real value of money widens the bid side of the book in real terms and narrows the ask side, so bids enter and asks exit until the probability of execution restores zero profit; a higher posting cost lowers quoting on both sides.

Because the two sides of the book generally differ in width, free entry delivers different quote intensities on the two sides of the market, although their real widths sum to the gain from trade,

[14] Jovanovic and Menkveld (2022) generate a similar dispersion of bids, but their model considers only the bid side of the market and abstracts from both monetary policy considerations and asset price determination.

$\varphi^m(\Gamma_a + \Gamma_b) = \varepsilon_H - \varepsilon_L$. Quoting is active on both sides, $\theta_a, \theta_b > 0$, only if the real width of each side of the book exceeds the posting cost, which requires $\kappa < (\varepsilon_H - \varepsilon_L)/2$.[15] The following corollary characterises the shape of the equilibrium price distributions.

Corollary 1. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then*

(i) The ask density $a(p)$ strictly decreases in p, is convex and prices are therefore right-skewed.

(ii) The bid density $b(p)$ strictly increases in p, is convex and prices are therefore left-skewed.

Because a higher ask earns a larger spread, brokers are indifferent across asks only if higher asks are less likely to execute. Asks are therefore posted less densely the further they are from the interbroker price, and bids likewise, which produces the convex and skewed shapes documented for order books (Section 7).

The properties of the ask and bid distributions extend naturally to their order-statistic counterparts, the best-ask and best-bid distributions, defined as the minimum and maximum of the k observed quotes, respectively.

Corollary 2. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then*

(i) There exists a unique best-ask distribution given by

$$\bar{A}(p) = \begin{cases} 1 - \frac{\kappa}{\varphi^m(p - \hat{p})} & \text{if } p \in \left[\hat{p} + \frac{\kappa}{\varphi^m}, \bar{p}\right) \\ 1 & \text{if } p = \bar{p}. \end{cases} \tag{15}$$

The corresponding best-ask density $\bar{a}(p)$ strictly decreases in p, is convex and prices are therefore right-skewed.

(ii) There exists a unique best-bid distribution given by

$$\bar{B}(p) = \begin{cases} 0 & \text{if } p = \underline{p} \\ \frac{\kappa}{\varphi^m(\hat{p} - p)} & \text{if } p \in \left(\underline{p}, \hat{p} - \frac{\kappa}{\varphi^m}\right]. \end{cases} \tag{16}$$

The corresponding best-bid density $\bar{b}(p)$ strictly increases in p, is convex and prices are therefore left-skewed.

[15] The threshold $\bar{\mu}$ defined below is also decreasing in κ, so for a given money growth rate the existence condition $\mu < \bar{\mu}$ in Proposition 2 holds only if κ is sufficiently small.

Figure 1 illustrates these distributions in a numerical example, which the remaining figures also use.[16] The book is not a parameter: its edges are investors' reservation prices, which lie on either side of the interbroker price. Posted quotes in panel (a) are densest just outside the interbroker price, and executable prices in panel (b), the best of the quotes an investor observes, are concentrated even more tightly around the inside of the book. The jumps in $\bar{B}(p)$ at $\underline{p}$ and in $\bar{A}(p)$ at $\bar{p}$ are mass points of size $e^{-\theta_b}$ and $e^{-\theta_a}$, the probabilities that a seller observes no bid and that a buyer observes no ask.[17] Section 5 interprets the first as the probability of a flash crash.

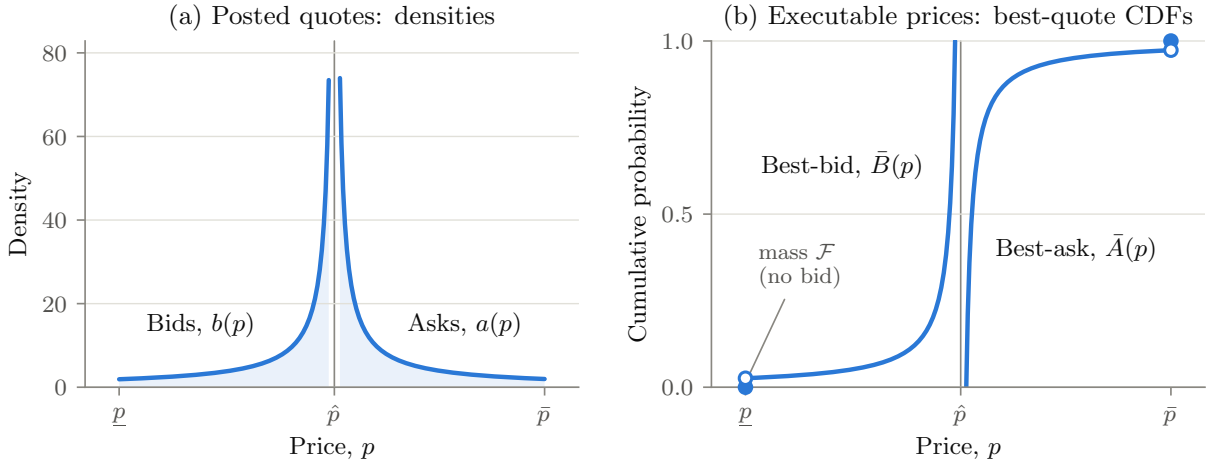


Figure 1: Equilibrium price dispersion

3.5 Existence and Uniqueness

The real value of money and the real equity price are determined jointly by investors' portfolio choices and brokers' quoting. It is convenient to define the upper bound on money growth for which a monetary equilibrium exists. Let $\hat{\phi}^s \equiv [\bar{\beta}\eta/(1-\bar{\beta}\eta)]\bar{\varepsilon}$ denote the fundamental

[16] The parameter values are $\beta = 0.96$, $\bar{\gamma} = 1$, $\mu = 1.02$, $\eta = 0.7$, $\pi_H = \pi_L = 0.5$, $\varepsilon_H = 1.5$, $\varepsilon_L = 0.5$, $\hat{p} = 1$ and $\kappa = 0.0131$. A period is a year, so $\mu = 1.02$ corresponds to 2 percent inflation. The dividend growth rate, the interbroker price and the mean valuation, $\bar{\varepsilon} = 1$, are normalisations, and η and the type probabilities are illustrative. The posting cost is set so that the mean number of quotes that an investor receives, $(\theta_a + \theta_b)/2$, is 3.64 at the baseline, with $\theta_b = 3.65$, $\theta_a = 3.63$ and $\varphi^m = 3.52$. This is the average number of dealer quotes that Riggs et al. (2020) report for customers who request quotes for index credit default swaps on the two largest dealer-to-customer swap execution facilities. The implied probability of receiving no bid, 2.6 percent, is below the 8 percent of their requests that do not end in a trade, a share that also includes requests in which the customer rejects every quote received. The welfare analysis in Section 6 also uses a resource cost of quoting a share of $\kappa_0 = 0.006$, so that the baseline regulatory wedge is $\tau = \kappa - \kappa_0 = 0.0071$. Figures 2–7 vary μ or κ around this baseline.

[17] As in Jovanovic and Menkveld (2022), $\bar{B}(p)$ is the probability that the best-bid is below p, so $\bar{B}(\underline{p}) = 0$, and the mass point is the jump of size $e^{-\theta_b}$ just above $\underline{p}$: a seller who receives no bid keeps the share, which is worth the same to them as selling at their reservation price $\underline{p}$.

value of equity (Section 4.3), let $\hat{v}_I \equiv \varepsilon_I + \hat{\phi}^s$ denote the value of a share to an investor of type $I \in \{H, L\}$ when the equity price equals its fundamental value, and

$$\bar{\mu} \equiv \bar{\beta}_L + \bar{\beta}_H \frac{\hat{v}_H}{\hat{v}_L + \kappa} \left\{ 1 - \frac{\kappa \left(\varepsilon_H - \varepsilon_L - \kappa \right)}{\left(\hat{v}_L + \kappa \right) \left(\varepsilon_H - \varepsilon_L - 2\kappa \right)} \ln \left[\frac{\left(\varepsilon_H - \varepsilon_L - \kappa \right) \left(\hat{v}_L + 2\kappa \right)}{\kappa \, \hat{v}_H} \right] \right\}. \tag{17}$$

As money growth approaches $\bar{\mu}$, bid quoting vanishes and the equity price falls to its fundamental value.

Proposition 2. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, $\kappa < (\varepsilon_H - \varepsilon_L)/2$ and $\bar{\beta} < \mu < \bar{\mu}$, then there exists a unique recursive monetary equilibrium in which the real (ex-dividend) equity price is*

$$\phi^s = \frac{\bar{\beta}_H \eta}{1 - \bar{\beta}_H \eta} \left\{ \varepsilon_H + \frac{\pi_L}{1 - \pi_L} \left[\varphi^m \hat{p} - \frac{\kappa \theta_b}{1 - e^{-\theta_b}} \right] \right\}, \tag{18}$$

and the real value of money φ^m is the unique solution to

$$\frac{\varepsilon_H + \phi^s}{1 - e^{-\theta_a}} \int_{\hat{p} + \frac{\kappa}{\varphi^m}}^{\bar{p}} \frac{\kappa}{p \left[\varphi^m \left(p - \hat{p} \right) \right]^2} \mathrm{d}p = \frac{\mu - \bar{\beta}_L}{\bar{\beta}_H}, \tag{19}$$

where $\bar{p} = (\varepsilon_H + \phi^s)/\varphi^m$, $\underline{p} = (\varepsilon_L + \phi^s)/\varphi^m$, and θ_a and θ_b are given by Proposition 1.

The equity price has two parts. The first, $[\bar{\beta}_H \eta/(1 - \bar{\beta}_H \eta)]\varepsilon_H$, is the value of the dividend stream to high-valuation investors. The second reflects the option to resell: a low-valuation investor can sell at the best-bid, whose real expected value conditional on trade is $\varphi^m \hat{p} - \kappa\theta_b/(1 - e^{-\theta_b})$, so the equity price depends on how actively brokers bid. The second condition determines the real value of money: its left-hand side is the liquidity return on money used to buy equity at the best-ask, conditional on trade, which equals $\mathbb{E}[\bar{p}/p \mid \text{trade}]$ because the buyer's reservation price $\bar{p}$ measures what a share is worth to the buyer, and its right-hand side is the opportunity cost of holding money, which rises with inflation. Higher inflation therefore requires buyers to trade further below their reservation price, which moves the real interbroker price towards sellers' reservation value: sellers' gain from trade at the interbroker price, $\varphi^m \Gamma_b$, falls, and buyers' gain, $\varphi^m \Gamma_a$, rises.[18]

[18] The expression for Z is $Z = \pi_L (1 - e^{-\theta_b}) \left[\pi_H \int_{\hat{p} + \kappa/\varphi^m}^{\bar{p}} \frac{\kappa}{p[\varphi^m (p - \hat{p})]^2} \, \mathrm{d}p \right]^{-1}$. This expression is omitted from the main text to maintain focus on the core equilibrium relationships. In the numerical example, Z falls with both μ and κ, so higher inflation or posting costs reduce the aggregate value of money balances held by investors.

4 Asset Prices

This section asks how inflation and posting costs affect the prices of the two assets, money and equity, and how much of the equity price reflects the option to resell. Both policies lower the real value of money and the real equity price, but they work differently. Inflation acts directly on the cost of holding money, whereas the posting cost acts only through the prices at which investors trade. Section 4.1 considers inflation, Section 4.2 the posting cost and Section 4.3 the speculative premium.

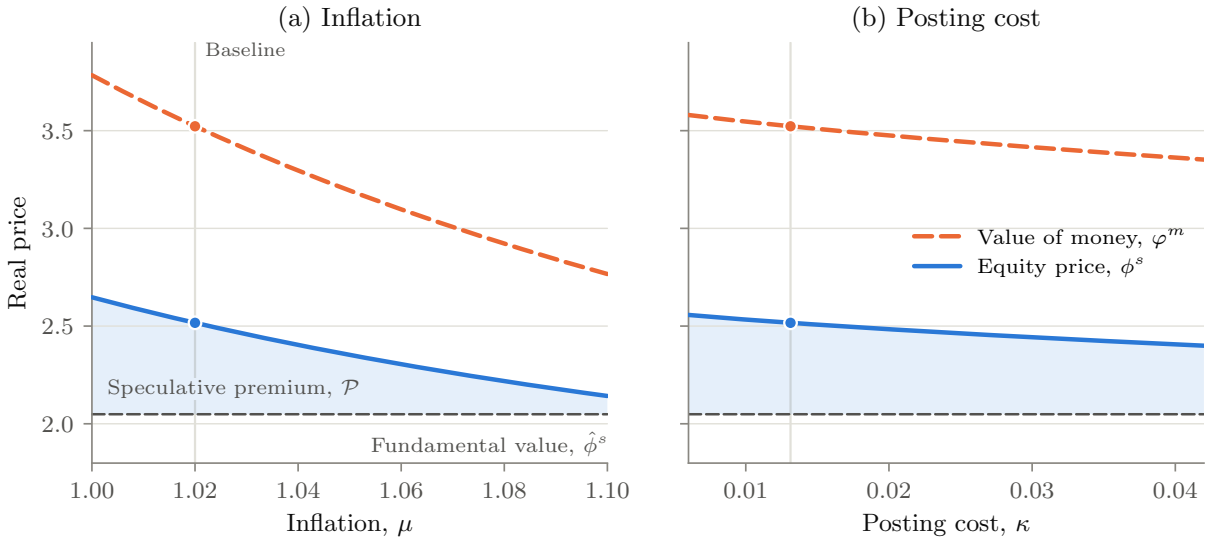


Figure 2: Asset prices, inflation and posting costs. The shaded gap between the equity price and its fundamental value is the speculative premium (Section 4.3); dots mark the baseline.

4.1 Inflation

Two conditions from Proposition 2 determine asset prices. By (19), investors hold money until its liquidity return, the expected ratio of a buyer's reservation price to the price paid, equals the cost of holding money, which rises with inflation. By (18), the equity price is the value of the dividend stream to high-valuation investors plus the value of the option to resell at the best-bid.

Inflation acts on both conditions. Higher inflation raises the cost of holding money between trading rounds, so investors hold money only if it earns a higher liquidity return. Buyers must therefore trade further below their reservation price, and the real value of money falls. The

real interbroker price falls with it, towards sellers' reservation value, so bids earn less and fewer brokers bid (Proposition 6). A seller can then expect a lower real price for the share, the option to resell loses value, and the real equity price falls.

Proposition 3. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The real value of money φ^m is decreasing with inflation μ, i.e., $\partial\varphi^m/\partial\mu < 0$.

(ii) The real (ex-dividend) price of the equity share ϕ^s is decreasing with inflation μ, i.e., $\partial\phi^s/\partial\mu < 0$.

Proposition 3 says that inflation lowers the real value of both assets. By the Fisher relation, $1+i = \mu/\bar{\beta}$, so a higher μ also means a higher nominal interest rate, and equity prices fall when nominal rates rise.[19] Panel (a) of Figure 2 illustrates: as inflation rises, the value of money and the equity price both fall, and the equity price approaches its fundamental value.

4.2 Posting Cost

A higher posting cost acts on the same two conditions, but only through quoting. Brokers post fewer quotes on both sides of the market (Proposition 6), so the best-ask that a buyer can expect rises and each unit of money buys less equity. The liquidity return on money falls, and investors hold money only if it is cheaper: the real value of money falls. A lower value of money and fewer bids also lower the real price that a seller can expect, so the option to resell loses value and the real equity price falls.

Proposition 4. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The real value of money φ^m is decreasing with the posting cost κ, i.e., $\partial\varphi^m/\partial\kappa < 0$.

(ii) The real (ex-dividend) price of the equity share ϕ^s is decreasing with the posting cost κ, i.e., $\partial\phi^s/\partial\kappa < 0$.

[19] The results of Proposition 3 are consistent with the resale-premium literature of Lagos and Zhang (2015, 2019a,b, 2020), where asset prices decline with inflation, making equity a poor hedge against inflation. This empirical relationship corresponds to what is commonly known as the 'Fed model'. For an extended discussion of the Fed model and its connection to resale-premium asset-pricing frameworks, see Lagos and Zhang (2019a).

Interpreting the regulation of intermediaries as an increase in κ, Proposition 4 implies that such regulation lowers asset prices. Panel (b) of Figure 2, drawn on the same scale as panel (a), shows that over the range plotted the posting cost moves asset prices far less than inflation does. The reason is that inflation acts directly on the cost of holding money, the right-hand side of (19), whereas the posting cost reaches asset prices only through the prices at which investors trade, and these lie within a narrow book around the interbroker price. The same cost matters far more for flash crash risk, as a comparison with panel (d) of Figure 3 shows.

4.3 Speculation

The equity price in Proposition 2 mixes two sources of value: the dividends that a share pays and the option to sell it. To separate them, it is standard to define the *fundamental value* of an asset as the expected present discounted value of its dividends, $\hat{\phi}^s_t \equiv \left[\bar{\beta}\eta/(1-\bar{\beta}\eta)\right]\bar{\varepsilon}y_t \equiv \hat{\phi}^s y_t$. Following Harrison and Kreps (1978), this paper defines the *speculative premium* as the difference between the market price and the fundamental value, $\mathcal{P}_t \equiv \phi^s_t - \hat{\phi}^s_t$. It measures the value that investors assign to the option to resell the asset: investors behave speculatively when this option leads them to pay more than they would if they had to hold the asset indefinitely.[20] Let $\mathcal{M}_b$ denote the mean best-bid conditional on trade, so that $\varphi^m\mathcal{M}_b = \varphi^m\hat{p} - \kappa\theta_b/(1-e^{-\theta_b})$ by Proposition 2. Then $\mathcal{P}_t = \mathcal{P}y_t$, where

$$\mathcal{P} = \frac{\bar{\beta}_H\eta}{1-\bar{\beta}_H\eta}\left[\varepsilon_H + \frac{\pi_L}{1-\pi_L}\varphi^m\mathcal{M}_b\right] - \hat{\phi}^s = \frac{\bar{\beta}_L\eta}{1-\bar{\beta}\eta}\left(\varphi^m\mathcal{M}_b - \varepsilon_L - \phi^s\right).$$

One might expect flash crashes, in which a seller cannot sell at all, to make the option to resell worth less than nothing. It never is. A seller who receives no bid keeps the share, which is exactly what the seller would have without the option, and a seller never accepts a bid below their reservation price. The real mean best-bid—the *cum-dividend* price that a seller can expect—therefore exceeds $\varepsilon_L + \phi^s$, the value of keeping the share, and the premium is positive. As in Lagos and Zhang (2015, 2019b), the resale option generates speculative premia.

Because the fundamental value does not depend on policy, every change in the equity price is a change in the premium. Higher inflation and higher posting costs both lower the real mean

[20] This paper's notion of *speculative premium* corresponds to the concept of a *speculative bubble* used in the modern literature, e.g., Barlevy (2007), Abreu and Brunnermeier (2003), Scheinkman and Xiong (2003), Scheinkman (2014), and Xiong (2013), following the *resale-option theory of bubbles* introduced by Harrison and Kreps (1978).

best-bid (Proposition 7 below), so the option to resell loses value and the premium falls.

Proposition 5. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The speculative premium $\mathcal{P}$ is decreasing with inflation μ, i.e., $\partial\mathcal{P}/\partial\mu < 0$.

(ii) The speculative premium $\mathcal{P}$ is decreasing with the posting cost κ, i.e., $\partial\mathcal{P}/\partial\kappa < 0$.

Proposition 5 implies that both policies push asset prices towards fundamentals, and that their whole effect on the equity price works through the option to resell. Regulation that raises the cost of market making, as discussed by Duffie (2012, 2018), therefore lowers equity valuations through the resale channel. In Figure 2, the premium is the shaded gap between the equity price and its fundamental value. It narrows sharply as inflation rises (panel (a)), but only modestly as the posting cost rises (panel (b)).

5 Price Dispersion and Flash Crashes

This section asks how inflation and posting costs change the terms on which investors trade: the prices at which they can buy and sell, the bid–ask spread and the risk of a flash crash. These results work through quote intensity, the mean number of asks that a buyer receives, θ_a, and the mean number of bids that a seller receives, θ_b. For executable prices, policy also matters through the edges of the book, which are investors' reservation prices. Section 5.1 shows how policy moves quote intensity, and Sections 5.2–5.4 trace the effects on executable prices, spreads and flash crashes.

5.1 Quote Intensity

A broker posts a quote only if the quote is expected to earn its cost. By Proposition 1, free entry sets quote intensity on side $j \in \{a, b\}$ of the market to $\theta_j = \ln(\varphi^m \Gamma_j/\kappa)$, the logarithm of the ratio of two numbers. The first, $\varphi^m\Gamma_j$, is the real width of that side of the book: the real spread that a quote at the edge of the book earns. The second is the posting cost. The real widths of the two sides add up to the gain from trade, $\varphi^m(\bar{p} - \underline{p}) = \varepsilon_H - \varepsilon_L$, and the real interbroker price, $\varphi^m\hat{p}$, determines how that gain is split between them.

The two policies act on different numbers. A higher posting cost raises the cost of quoting on both sides of the market, so brokers post fewer asks and fewer bids. Inflation instead changes

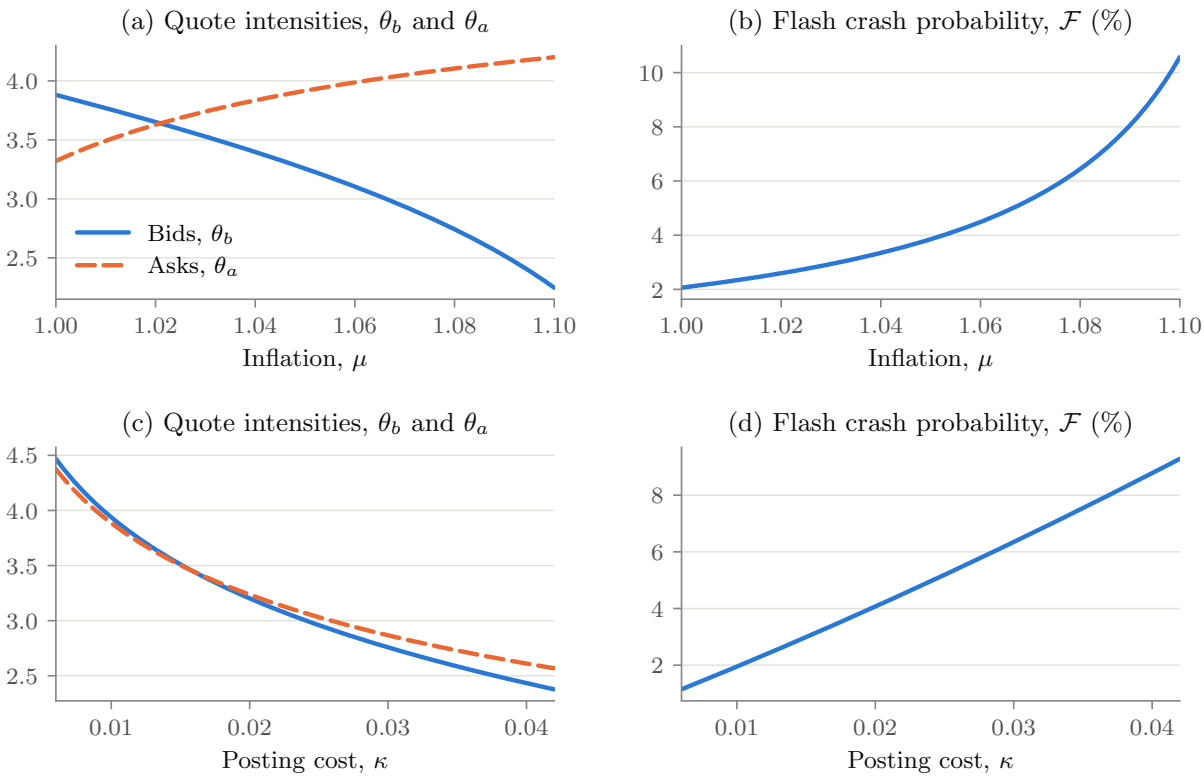


Figure 3: Quote intensity and the flash crash probability

the split. Higher inflation raises the cost of carrying money between trading rounds and lowers the real value of money, so the real interbroker price falls relative to investors' valuations of the equity. A buyer then gains more from buying at the interbroker price, and a seller gains less from selling at it: asks earn more and bids less, so more brokers ask and fewer bid.

Proposition 6. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The bid intensity θ_b is decreasing, and the ask intensity θ_a increasing, with inflation μ, i.e., $\partial\theta_b/\partial\mu < 0 < \partial\theta_a/\partial\mu$.

(ii) The quote intensities θ_a and θ_b are decreasing with the posting cost κ, i.e., $\partial\theta_a/\partial\kappa < 0$ and $\partial\theta_b/\partial\kappa < 0$.

Proposition 6 shows that both policies drive bids out of the market, but they differ on the ask side: inflation draws asks in, whereas a higher posting cost drives them out. Figure 3 illustrates. As inflation rises, bid intensity falls and ask intensity rises, and the two curves cross

where the two sides of the book are equally wide (panel (a)). As the posting cost rises, both fall (panel (c)). Panels (b) and (d) show the implied flash crash probability, $\mathcal{F} = e^{-\theta_b}$, which Section 5.4 studies.

5.2 Executable Prices

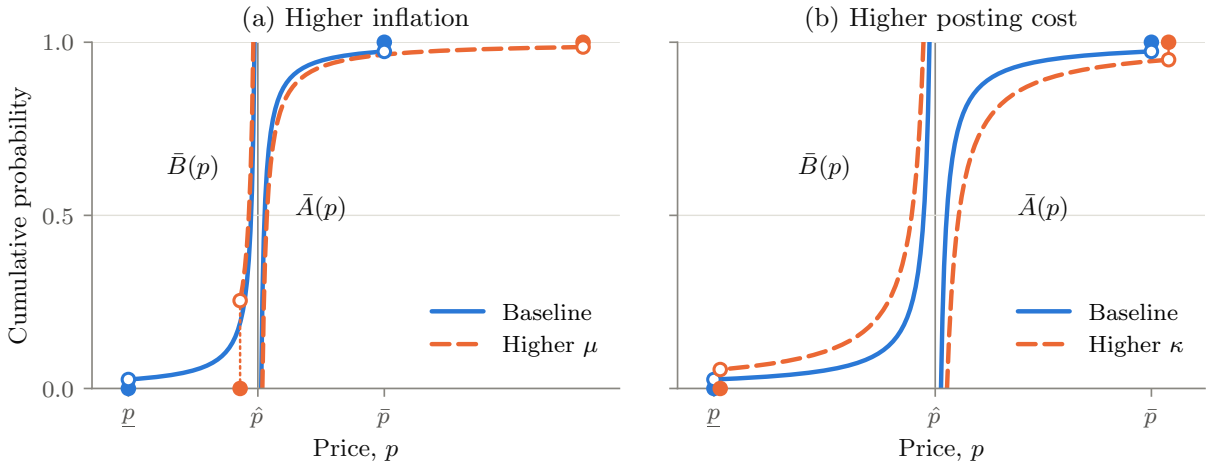


Figure 4: Inflation, posting costs and the distribution of executable prices

Quote intensity matters to investors because it determines the prices at which they trade. Because $\kappa/\varphi^m = \Gamma_a e^{-\theta_a} = \Gamma_b e^{-\theta_b}$ (Proposition 1), the best-quote distributions in Corollary 2 can be written, on the continuous parts of their supports, as

$$\bar{A}(p) = 1 - \frac{\Gamma_a e^{-\theta_a}}{p - \hat{p}} \quad \text{and} \quad \bar{B}(p) = \frac{\Gamma_b e^{-\theta_b}}{\hat{p} - p}. \tag{20}$$

Given the book $[\underline{p}, \bar{p}]$, quote intensity is therefore a sufficient statistic for the distribution of executable prices: θ_a for asks and θ_b for bids. Let $\mathcal{M}_a$ denote the mean best-ask conditional on trade, and $\mathcal{V}_a$ and $\mathcal{V}_b$ the variances of the best-ask and the best-bid conditional on trade. By (20),

$$\mathcal{M}_a - \hat{p} = \frac{\Gamma_a \theta_a}{e^{\theta_a} - 1} \quad \text{and} \quad \mathcal{V}_a = \Gamma_a^2 \left[e^{-\theta_a} - \frac{\theta_a^2}{(e^{\theta_a} - 1)^2} \right],$$

and $\hat{p} - \mathcal{M}_b$ and $\mathcal{V}_b$ are the same functions of Γ_b and θ_b.

One might expect that, because inflation draws asks into the market, buyers would pay a smaller markup over the interbroker price. They pay a larger one. The posting cost is fixed in

units of the consumption good, so in units of money quoting a share costs κ/φ^m, and no broker posts an ask below $\hat{p} + \kappa/\varphi^m$. By lowering the value of money, inflation raises the money cost of quoting and pushes every executable ask up. The additional asks only lower the probability that a buyer receives none and is left at the top of the book, $e^{-\theta_a}$. A higher posting cost raises the cost of quoting directly and also removes asks, so executable asks rise for both reasons.

The dispersion of executable asks behaves differently. Given the book, the mean best-ask moves monotonically with quote intensity, but the variance does not: $e^{-\theta} - \theta^2/(e^\theta - 1)^2$ increases with θ for $\theta < \bar{\theta}$ and decreases for $\theta > \bar{\theta}$, where $\bar{\theta} \approx 1.73$. In an active market, $\theta_a > \bar{\theta}$, fewer asks spread executable asks out. In a very thin market, executable asks are already concentrated near the top of the book, and fewer asks concentrate them further. On the bid side, fewer bids move executable bids away from the interbroker price, but inflation also narrows the bid side of the book, so its effect on nominal bids is ambiguous. In real terms, sellers receive less.

Proposition 7. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) An increase in inflation μ or in the posting cost κ shifts the best-ask distribution up, and the distribution of the real best-bid, $\varphi^m p$, down, in the sense of first-order stochastic dominance.

(ii) The mean best-ask $\mathcal{M}_a$ is increasing, and the real mean best-bid $\varphi^m \mathcal{M}_b$ is decreasing, with inflation μ and with the posting cost κ.

(iii) The variance of the best-ask, $\mathcal{V}_a$, is increasing with inflation μ and, if $\theta_a > \bar{\theta}$, with the posting cost κ.

Figure 4 illustrates part *(i)* for higher inflation and for a higher posting cost, with prices measured in money. Both lower $\bar{A}(p)$ at every price, so executable asks move up, and both enlarge the mass point at the bottom of the book, the probability of a flash crash. On the bid side, the two panels differ. A higher posting cost barely moves the book, and executable bids move down (panel (b)). Inflation, by contrast, moves the book itself. Because it lowers the value of money, it raises investors' reservation prices measured in money, and the bid side of the book narrows towards the interbroker price (panel (a)). Nominal bids at the bottom of the book therefore rise, and the two best-bid distributions in panel (a) cross: in money terms, neither

dominates the other. Part *(i)* ranks bids in real terms. Measured in goods, every quantile of the best-bid falls (Corollary 3), because the value of money falls by more than nominal bids rise.[21]

5.3 Spreads

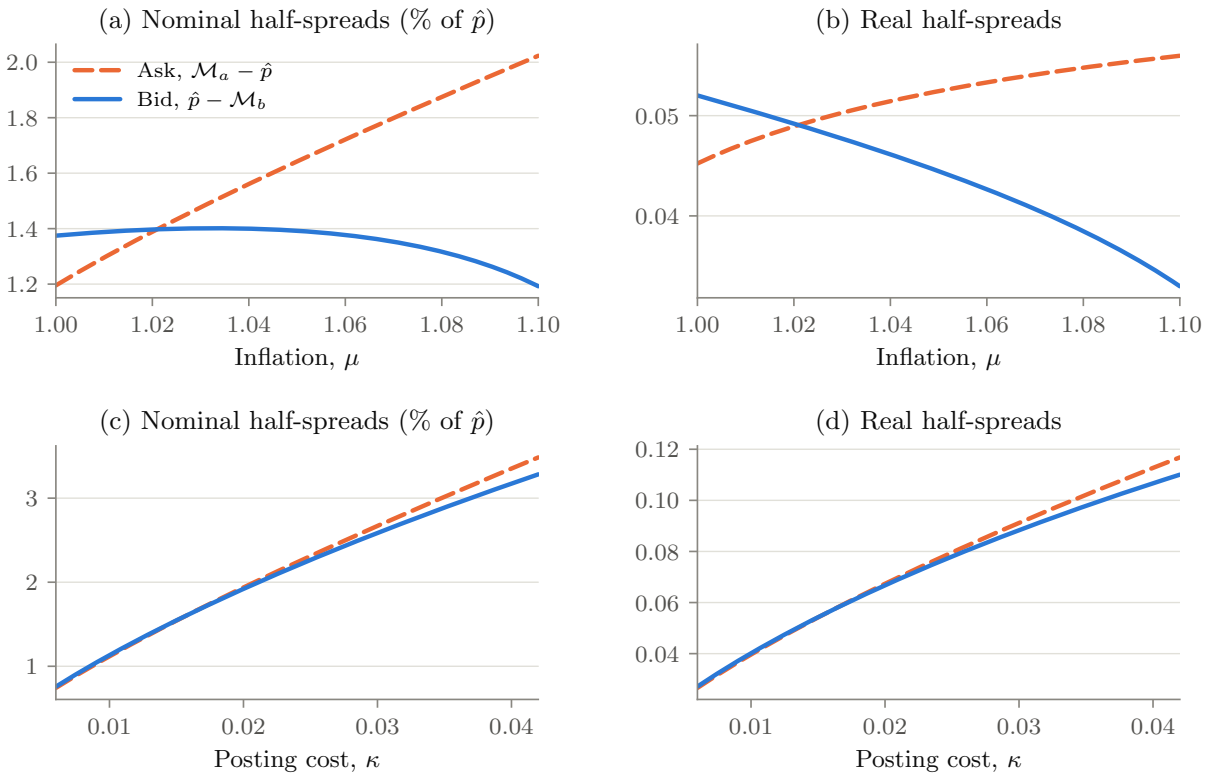


Figure 5: Nominal and real bid–ask spreads

The *nominal bid–ask spread* is the distance between the mean best-ask and the mean best-bid, $\mathcal{S}^m \equiv \mathcal{M}_a - \mathcal{M}_b$, the sum of the ask half-spread, $\mathcal{M}_a - \hat{p}$, and the bid half-spread, $\hat{p} - \mathcal{M}_b$. The *real bid–ask spread*,

$$\mathcal{S} \equiv \varphi^m \mathcal{S}^m = \frac{\kappa\theta_a}{1 - e^{-\theta_a}} + \frac{\kappa\theta_b}{1 - e^{-\theta_b}},$$

measures the cost of a round trip in units of the consumption good. Each term is a real half-spread per share: the cost per share of the quotes that an investor receives on average, $\kappa\theta_j$, divided by the probability of trade, $1 - e^{-\theta_j}$. The reason is free entry. Brokers earn zero

[21] The distribution of posted asks, $A(p)$, shifts in the same way.

expected profit, so the investors who trade pay, through the spread, for every quote that is posted, including those that never execute. A higher posting cost raises both terms directly, while the induced fall in quote intensity lowers them. Inflation moves the two terms in opposite directions.

Proposition 8. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The nominal ask half-spread, $\mathcal{M}_a - \hat{p}$, is increasing with inflation μ and with the posting cost κ. Inflation raises the real ask half-spread, $\kappa\theta_a/(1-e^{-\theta_a})$, and lowers the real bid half-spread, $\kappa\theta_b/(1-e^{-\theta_b})$.

(ii) Let $\epsilon_{b,\kappa} \equiv \partial \ln(\varphi^m \Gamma_b)/\partial \ln \kappa$ denote the elasticity of the real width of the bid side of the book with respect to the posting cost. If $|\epsilon_{b,\kappa}| \leq \min\{1, \Gamma_a/\Gamma_b\}$, the nominal and real bid–ask spreads are increasing with the posting cost κ, i.e., $\partial \mathcal{S}^m/\partial\kappa > 0$ and $\partial \mathcal{S}/\partial\kappa > 0$.

Proposition 8 shows that the two policies affect spreads differently. Inflation shifts the cost of intermediation from sellers to buyers: it widens the real half-spread that buyers pay and narrows the one that sellers pay, because the real interbroker price falls towards sellers' reservation value. The response of the whole spread to monetary policy is therefore ambiguous, as in the monetary resale literature (e.g., Lagos and Zhang, 2015). A higher posting cost raises the spread, provided that the real width of the bid side of the book does not respond too strongly to it, which is the condition in part *(ii)*. The condition holds throughout the numerical example.[22]

Figure 5 illustrates. As inflation rises, the real ask half-spread rises and the real bid half-spread falls (panel (b)), so their sum, the real bid–ask spread, changes much less than either. As the posting cost rises, both real half-spreads rise steeply (panel (d)). Panels (a) and (c) show the nominal half-spreads, as percentages of the interbroker price.

5.4 Flash Crashes and Tail Risk

A *flash crash* is the event in which a seller receives no bid and so cannot sell. Its probability is $\mathcal{F} = P_0(\theta_b) = e^{-\theta_b}$. As in Jovanovic and Menkveld (2022), flash crashes arise from brokers' decisions rather than from shocks: when the expected spread of a bid is small relative to the

[22] The real half-spread has the same form as the real spread in Jovanovic and Menkveld (2022), and so responds to the posting cost in a similar way: its direct effect outweighs the offsetting effect of lower quote intensity under the condition in Proposition 8*(ii)*. Their model has only the bid side of the market and no money.

cost of posting it, brokers post few bids, and because bids reach sellers at random, some sellers receive none. Liquidity can therefore vanish without any shock to fundamentals or informational friction.

Proposition 9. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The flash crash probability $\mathcal{F}$ is increasing with inflation μ, i.e., $\partial\mathcal{F}/\partial\mu > 0$.

(ii) The flash crash probability $\mathcal{F}$ is increasing with the posting cost κ, i.e., $\partial\mathcal{F}/\partial\kappa > 0$.

Proposition 9 has a direct implication for regulation. Regulation intended to make intermediaries safer can make markets more fragile: by raising the cost of market making, it raises the probability that no bid arrives. Higher inflation has the same effect. Panels (b) and (d) of Figure 3 show both: the flash crash probability rises with the posting cost, and it rises with inflation at an increasing rate.

A flash crash is the most extreme event in the tail of executable bids, and its probability governs the rest of that tail. To make this precise, let $\mathcal{Q}_a(z) \equiv \inf\{p : \bar{A}(p) \geq z\}$ and $\mathcal{Q}_b(z) \equiv \inf\{p : \bar{B}(p) \geq z\}$ denote the quantile functions of the best-ask and best-bid distributions. They measure the Value-at-Risk of executable prices: with probability at least z a buyer pays no more than $\mathcal{Q}_a(z)$, and with probability at least $1 - z$ a seller receives no less than $\mathcal{Q}_b(z)$.

Corollary 3. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The Value-at-Risk (quantile function) of the best-ask distribution is

$$\mathcal{Q}_a(z) = \begin{cases} \hat{p} + \dfrac{\kappa}{\varphi^m(1-z)} & \text{for } z \in [0, 1 - e^{-\theta_a}), \\ \bar{p} & \text{for } z \in [1 - e^{-\theta_a}, 1]. \end{cases}$$

(ii) The Value-at-Risk (quantile function) of the best-bid distribution is

$$\mathcal{Q}_b(z) = \begin{cases} \underline{p} & \text{for } z \in [0, \mathcal{F}], \\ \hat{p} - \dfrac{\kappa}{\varphi^m z} & \text{for } z \in (\mathcal{F}, 1]. \end{cases}$$

(iii) An increase in inflation μ or in the posting cost κ raises $\mathcal{Q}_a(z)$ at every z, strictly whenever $\mathcal{Q}_a(z) < \bar{p}$ before the increase, lowers the real value of the best-bid quantile, $\varphi^m \mathcal{Q}_b(z)$, at every z, and lengthens the segment on which $\mathcal{Q}_b(z) = \underline{p}$.

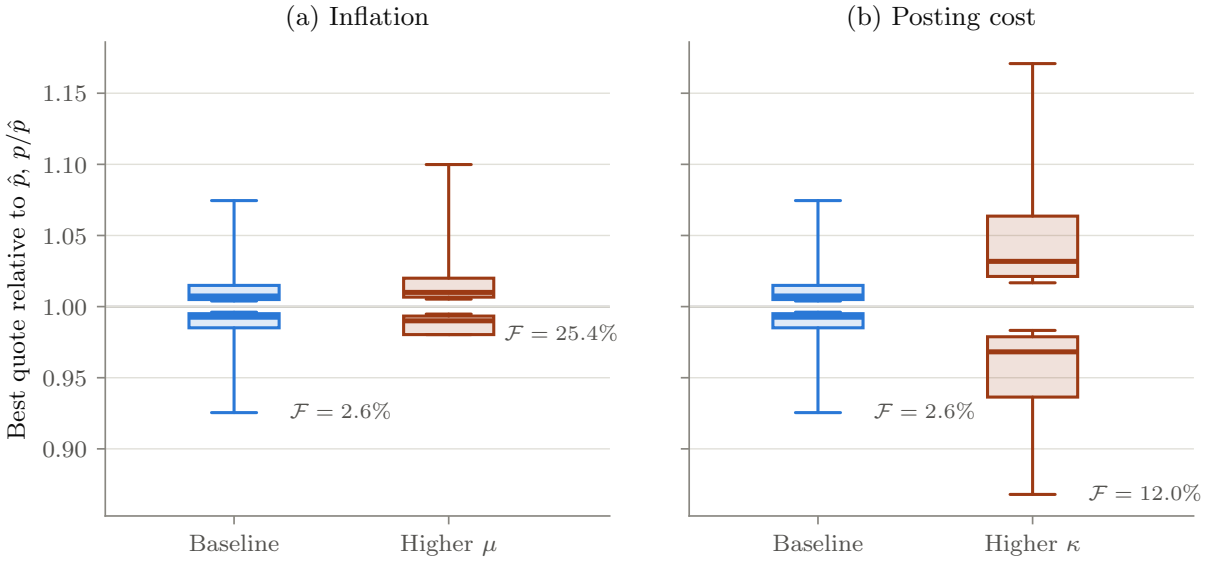


Figure 6: Value-at-Risk: the distribution of executable prices and its tails

Corollary 3 shows that the quantile function of the best-bid has a flat segment of length $\mathcal{F}$ at the bottom of the book: with probability $\mathcal{F}$ a seller receives no bid, and the corresponding tail of the distribution sits at the worst price in the book, the seller's reservation price. The rest of the function is pinned down by $\kappa/\varphi^m = \Gamma_b \mathcal{F}$: the shortfall of the best-bid below the interbroker price that is exceeded with probability z equals $\Gamma_b \min\{1, \mathcal{F}/z\}$. Given the book, every quantile therefore moves away from the interbroker price as $\mathcal{F}$ rises, and at any confidence level with $z \leq \mathcal{F}$, the Value-at-Risk of a seller is the bottom of the book. Tail risk for sellers is thus governed by the flash crash probability. So is pricing error in normal times: conditional on receiving a bid, the mean-squared error of the best-bid around the interbroker price equals $\Gamma_b^2 \mathcal{F}$. Buyers face the same structure, with $e^{-\theta_a}$ in place of $\mathcal{F}$ and Γ_a in place of Γ_b.

Figure 6 summarises how the distributions of executable prices and their tails change. Following the box plots in Jovanovic and Menkveld (2022), it measures executable prices relative to the interbroker price, $p/\hat{p}$, so that the cases are directly comparable. In each case the upper box shows the best-ask and the lower box the best-bid; each box spans the interquartile range, the line inside marks the median, and the whiskers extend to the 5th and 95th percentiles.

Each panel compares the baseline with an extreme case: much higher inflation in panel (a) and a much higher posting cost in panel (b). A higher posting cost raises $\mathcal{F}$ and stretches both distributions away from the interbroker price, so that the whiskers reach the edges of the book. Higher inflation also raises $\mathcal{F}$ and stretches the ask side. One might expect it to stretch the bid side too. Instead, the bid side of the book narrows, because sellers' reservation price rises towards the interbroker price, and the bid box now ends at the bottom of the book. A flash crash therefore becomes more likely, but it leaves a seller's price closer to the interbroker price.

6 Optimal Monetary and Macroprudential Policy

This paper now evaluates policy by its effect on welfare. Because preferences are linear, spreads, asset prices, speculative premia and the inflation tax are transfers between agents and do not affect welfare directly. Welfare depends on two margins only: who holds the equity when dividends are paid, and the resources used to post quotes, which depend on how many quotes brokers post and how many shares those quotes cover.

Every investor enters the first subperiod holding A^s units of equity. A low-valuation investor who receives a bid sells the equity to a high-valuation investor (Lemma 1), but with probability $\mathcal{F} = e^{-\theta_b}$ receives no bid and keeps it, so a fraction $\pi_L \mathcal{F}$ of the equity stock stays with low-valuation investors. To separate the resource cost of intermediation from regulation, write the posting cost as $\kappa = \kappa_0 + \tau$, where $\kappa_0 > 0$ is the resource cost of quoting a share and $\tau \geq 0$ is a regulatory wedge, such as the private cost of capital or liquidity requirements on market makers, that raises brokers' cost of quoting without using up resources; any payment it involves is rebated lump sum to investors.[23] The comparative statics with respect to κ in Sections 4 and 5 apply to τ. A low-valuation investor receives θ_b bids on average, each covering the investor's A^s shares, and the asks that high-valuation investors receive cover ω shares for each of the $\pi_L(1 - e^{-\theta_b})A^s$ shares traded, where $\omega \geq 1$ because only the best ask that a buyer receives executes. Steady-state welfare per period, normalised by the aggregate dividend $y_t A^s$, is therefore

$$\mathcal{W} = \varepsilon_H - \pi_L(\varepsilon_H - \varepsilon_L)e^{-\theta_b} - \kappa_0 \pi_L \left[\theta_b + \omega\left(1 - e^{-\theta_b}\right)\right]. \tag{21}$$

[23] Treating the regulatory wedge as a resource cost would add a direct welfare cost of regulation and strengthen the conclusions below.

The first two terms are the dividend value of the equity stock net of the misallocation caused by flash crashes, and the last is the resource cost of quoting. Asks that do not execute use resources without changing who holds the equity, and asks cover only the shares traded, $\omega = 1$, only in the limit in which ask quoting vanishes. Let $\theta_b(\mu, \tau)$ denote equilibrium bid intensity, and $\theta_b(\bar{\beta}, \tau)$ its limit at the Friedman rule, $\mu \to \bar{\beta}$.

Proposition 10. *If $P_k(\theta)$ is Poisson with mean $\theta \in \mathbb{R}_+$ and $\varepsilon_H - \varepsilon_L > 2\kappa_0$, then in the recursive monetary equilibrium:*

(i) Welfare is maximised when asks cover only the shares traded, $\omega = 1$, and bid intensity is $\theta_b^* = \ln[(\varepsilon_H - \varepsilon_L - \kappa_0)/\kappa_0]$, so the efficient flash crash probability is $\mathcal{F}^* = \kappa_0/(\varepsilon_H - \varepsilon_L - \kappa_0) > 0$.

(ii) Equilibrium bid quoting is never excessive: $\theta_b \leq \theta_b^*$, with equality only in the limit at the Friedman rule without regulation, $\mu \to \bar{\beta}$ and $\tau = 0$.

(iii) Welfare is lower at every $\mu > \bar{\beta}$ than at the Friedman rule, which is therefore optimal.

(iv) Under optimal monetary policy, the flash crash probability is $(\kappa_0 + \tau)/(\varepsilon_H - \varepsilon_L - \kappa_0 - \tau)$, which equals $\mathcal{F}^*$ if $\tau = 0$ and exceeds it if $\tau > 0$, and welfare is strictly decreasing in τ.

Part *(i)* says that some flash crash risk is efficient. An additional bid to a low-valuation investor averts a misallocation worth $\varepsilon_H - \varepsilon_L$ per share with probability $e^{-\theta_b}$, the probability that the investor would otherwise receive no bid, but the trade it creates also needs an ask, which costs κ_0 per share, so the trade is worth $\varepsilon_H - \varepsilon_L - \kappa_0$ to society. The planner stops adding bids when this expected benefit equals the resource cost of quoting a share, κ_0. Asks, by contrast, only determine which buyers absorb the equity that sellers sell, so the planner would post only as many as trades require. Brokers make the same calculation for bids with different numbers. Free entry requires $e^{-\theta_b}\varphi^m\Gamma_b = \kappa$ (Proposition 1): a bid at the bottom of the book earns the real spread $\varphi^m\Gamma_b$ only when the seller has no other bid, and brokers add bids until this expected revenue equals the private cost κ. Part *(ii)* follows. The spread at the bottom of the book is the seller's share of the gain from trade at the interbroker price, and the buyer's share, $\varphi^m\Gamma_a$, is at least the cost of an ask, κ, so $\varphi^m\Gamma_b \leq \varepsilon_H - \varepsilon_L - \kappa$. A bid therefore never earns more than the trade it creates is worth to society, and it earns less whenever buyers receive more than

the cost of an ask or the cost of quoting includes a regulatory wedge. On how monetary policy affects the efficiency of OTC trade in a different setting, see Geromichalos and Jung (2019).

Lower inflation raises the real spread at the bottom of the book, $\varphi^m \Gamma_b$, and with it bid quoting, and higher inflation lowers them (Proposition 6). At the Friedman rule, money earns no liquidity return, so buyers pay their reservation price, ask quoting vanishes, and asks cover only the shares traded. Sellers then receive the whole gain from trade at the interbroker price, net of the cost of an ask: $\varphi^m \Gamma_b = \varepsilon_H - \varepsilon_L - \kappa$, and $\mathcal{F} = \kappa/(\varepsilon_H - \varepsilon_L - \kappa)$. Without regulation, $\kappa = \kappa_0$, so a bid earns exactly what the trade it creates is worth to society, and the Friedman rule brings bid quoting, and with it the flash crash probability, to their efficient levels. Higher inflation shifts the gain from trade from sellers to buyers, which pushes bid quoting below its efficient level and draws in asks that do not execute, so welfare is lower at every money growth rate above the Friedman rule (part *(iii)*). Part *(iv)* describes the interaction with regulation. A regulatory wedge lowers bid quoting just as inflation does, and monetary policy, already at the Friedman rule, cannot lower inflation any further to counteract it: with any regulatory wedge, flash crashes are more likely than is efficient, and tighter regulation lowers welfare further. High inflation combined with tight regulation is therefore doubly costly, because both push bid quoting further below its efficient level.

Figure 7 illustrates in the numerical example, in which the baseline posting cost is split into a resource cost κ_0 and a regulatory wedge τ. Panel (a) plots the flash crash probability against money growth without regulation and for a loose, the baseline and a tight regulatory wedge, and dots mark the Friedman rule. Without regulation, the flash crash probability falls to its efficient level $\mathcal{F}^*$ at the Friedman rule. Under every wedge it stays above $\mathcal{F}^*$, even at the Friedman rule, and the tighter the regulation, the further above it the curve lies. Panel (b) plots the welfare loss relative to the efficient allocation. It rises with money growth in every case, so the Friedman rule is optimal. At the Friedman rule, the loss vanishes without regulation and grows with the wedge.

The model abstracts from the benefits of regulating intermediaries, such as limiting their risk-taking, which would justify a positive wedge. What it adds is the cost side: by raising the private cost of quoting, such regulation lowers quote intensity and raises the probability of flash crashes, a cost that monetary policy cannot reduce once inflation is at the Friedman rule.

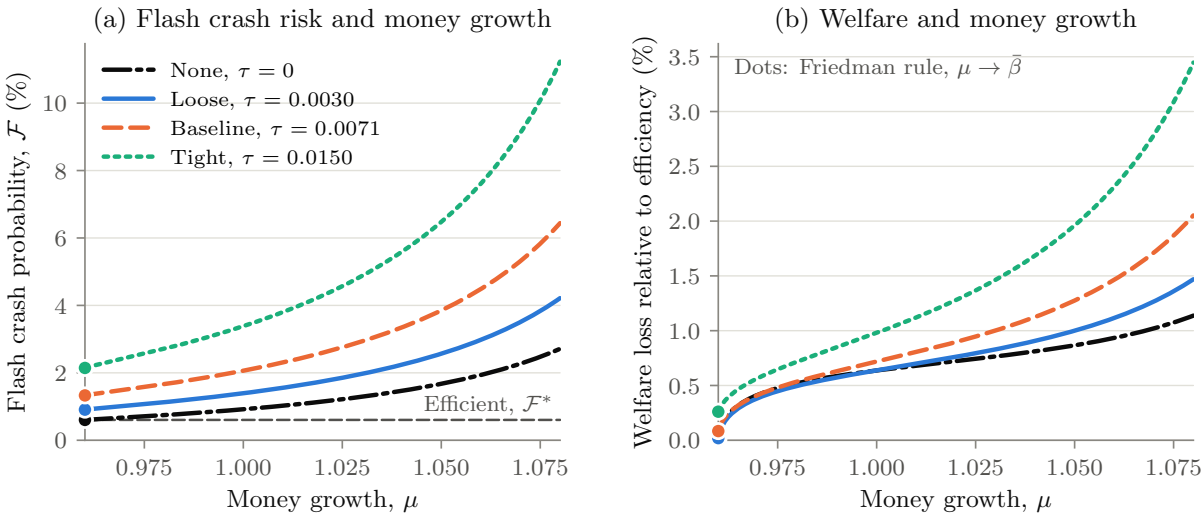


Figure 7: Flash crash risk and optimal monetary policy under regulation

7 Discussion

This section relates the model to five features of the data. For each feature, it identifies the counterpart of the empirical object in the model and asks whether the model reproduces the pattern qualitatively. The model matches the shape of quote distributions, the decline in intermediary participation, the fall in depth relative to spreads, the occasional disappearance of liquidity and the co-movement of valuations and trading activity.

First, the shape of quote distributions. Across countries and asset classes, bids cluster below the mid-price and asks above it, and quote densities are asymmetric and convex (Biais et al., 1995; Goldstein and Kavajecz, 2004; Hollifield et al., 2006; Næs and Skjeltorp, 2006; Degryse et al., 2015). In the model, the densities of posted quotes are convex and skewed away from the interbroker price on both sides of the market (Proposition 1 and Corollary 1), and their tails are long, as are those of the limit-order prices documented by Zovko and Farmer (2002), Weber and Rosenow (2005) and Gu et al. (2008). Executable prices are more concentrated still: the best-quote distributions of Corollary 2 put most of their mass near the inside of the book.

Second, the decline in intermediary participation. Jovanovic and Menkveld (2022) estimate that the average number of middlemen bidding for SPY, the exchange-traded fund that tracks the S&P 500, fell from about 50 to fewer than 25 between 2007 and 2018, with the lion's share of the decline around the implementation of the Volcker Rule in 2013–2014, and dealers increasingly avoid acting as principal (Bessembinder et al., 2018; Choi et al., 2024). Their

estimated number of bidders is the empirical counterpart of bid intensity: in their model, the number of bids is Poisson with a mean equal to the logarithm of the ratio of the value of winning, net of the seller's reservation value, to the cost of bidding, which is the same form as $\theta_b = \ln(\varphi^m \Gamma_b/\kappa)$ (Proposition 1). They find that the cost of bidding is more or less constant over their sample, so the decline reflects a fall in the value of winning. Since θ_b depends only on the ratio of the two, a higher posting cost and a narrower real bid side of the book, $\varphi^m \Gamma_b$, have the same effect. In the model, higher inflation also narrows the real bid side of the book (Proposition 6). Because the probability of a flash crash depends only on bid intensity, inflation and a higher posting cost that reduce bid intensity by the same amount raise flash crash risk by the same amount, but in the numerical example inflation lowers the equity price by more. Jovanovic and Menkveld (2022) also note other possible explanations for the decline, such as lower volatility and volume and the spread of electronic trading.

Third, depth has fallen by more than spreads. In the SPY sample of Jovanovic and Menkveld (2022), the bid–ask spread fell by 6.6 percent per year, while from 2011 onwards depth at the best quote fell by 19.1 percent per year and depth within 50 basis points of the mid-price by 27.0 percent per year. The counterpart of depth in the model is quote intensity. Lower bid intensity moves executable bids away from the interbroker price, given the book, but higher money growth also narrows the bid side of the book and lowers the real bid half-spread (Propositions 7 and 8), so spreads need not widen as depth falls. Because tail risk is governed by the probability of a flash crash rather than by the average spread (Corollary 3), the spread alone is an incomplete measure of market conditions.

Fourth, liquidity occasionally vanishes. Jovanovic and Menkveld (2022) estimate that the probability of an empty order book for SPY averaged 0.018 percent which, treating each day as an independent draw, implies a 40 percent chance of at least one empty book in their 11.5-year sample. During the flash crash of 6 May 2010, prices of several U.S. equities collapsed to near zero before recovering within minutes, consistent with a brief absence of liquidity rather than a change in fundamentals (Menkveld and Yueshen, 2019). In the model, such episodes arise without any shock to fundamentals and become more likely as bid intensity falls (Proposition 9 and Corollary 3).

Fifth, speculative episodes combine high valuations with heavy trading (Scheinkman and Xiong, 2003; Scheinkman, 2014). In the model, bid intensity, a proxy for trading activity, and

the speculative premium move together whether the variation comes from inflation or from posting costs (Propositions 5 and 6): lower inflation raises both, and so does a lower posting cost.

8 Conclusion

This paper introduces price posting by brokers into a search-theoretic model of money and asset exchange and studies how monetary policy and the regulation of intermediaries affect asset price dispersion and flash crash risk. Both policies act through a single channel, quote intensity. Higher inflation and tighter regulation reduce the number of bids, lower equity prices and make flash crashes more likely. Surprisingly, inflation also draws in more asks even as it drives out bids, because it shifts the gain from trade at the interbroker price from sellers to buyers. The two policies also reshape the bid side of the book differently. Tighter regulation spreads executable bids towards the bottom of the book, whereas inflation also raises sellers' reservation price towards the interbroker price, so that a flash crash, although more likely, leaves a seller's price closer to the interbroker price.

The welfare-maximising probability of a flash crash is positive, because quotes are costly. Inflation shifts the gains from trade from sellers to buyers, so too few brokers bid, and the Friedman rule is optimal: without regulation, it brings flash crash risk down to its welfare-maximising level. With any regulation of intermediaries, however, flash crash risk remains above that level even at the Friedman rule. This suggests that part of the fragility of intermediated markets is a cost of regulating them: whenever intermediaries are regulated, even the best monetary policy leaves sellers too exposed to the risk of receiving no bid.

This paper has focused on a stylised market, in which brokers are identical and a posting cost stands in for regulation, in order to isolate how policy acts through quote intensity. This delivers a strong result: without any heterogeneity among brokers and without any shock to fundamentals, inflation and regulation alone change the probability that liquidity vanishes. The model also abstracts from the benefits of regulating intermediaries, such as limiting their risk-taking, and so captures only the cost side of regulation.

In future work, it would be interesting to explore how credit conditions and leverage shape the transmission of monetary and regulatory policy to asset prices, particularly through margin lending and short selling. Another direction for future research would be to take the model

to data on quote intensity and execution prices. How much of the decline in the number of intermediaries since the financial crisis reflects regulation rather than monetary policy, and how large are the resulting welfare costs? These questions are left for future research.

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Extension: Overdispersed Quote Arrivals

The Poisson assumption ties the variance of the number of quotes to its mean. In the data, quoting is far more uneven. The mean number of bidders that Jovanovic and Menkveld (2022) estimate for SPY falls from about 50 to fewer than 25 over their sample, yet the average probability of an empty book that they report, 0.018 percent, is at least seven orders of magnitude larger than e^{-m} for any mean number of bidders m between 25 and 50: empty books are far more common than a Poisson model with a constant mean of that many bidders allows. This extension replaces the Poisson distribution with the negative binomial, the standard model of overdispersed counts, and asks three questions. What does overdispersion change (Section E.1)? Is reducing it a useful policy lever (Section E.2)? And how far should it be reduced (Section E.3)?

Assumption 2. *The number of asks $k \in \{0, 1, 2, \dots\}$ received by a high-valuation investor and the number of bids received by a low-valuation investor are negative binomial with means $\theta_a \in \mathbb{R}_+$ and $\theta_b \in \mathbb{R}_+$ and a common dispersion parameter $r > 0$. That is, the probability of receiving k quotes on side $j \in \{a, b\}$ is given by*

$$P_k(\theta_j; r) = \binom{k + r - 1}{k} \left(\frac{r}{r + \theta_j}\right)^r \left(\frac{\theta_j}{r + \theta_j}\right)^k. \tag{22}$$

The variance of the number of quotes is $\theta_j + \theta_j^2/r$, so $1/r$ measures overdispersion, and the Poisson is the limit $r \to \infty$. The negative binomial has a simple interpretation. It arises when an investor receives a Poisson number of quotes with mean $\theta_j \psi$, where ψ is gamma distributed across investors with mean one and variance $1/r$: some investors are well connected and receive many quotes, others few. By (4), an ask at price p executes with probability $[1 + \theta_a A(p)/r]^{-(r+1)}$, which converges to $e^{-\theta_a A(p)}$ as $r \to \infty$, and a bid at price p executes with probability $[1 + \theta_b(1 - B(p))/r]^{-(r+1)}$. The probability of receiving no bid is $(1 + \theta_b/r)^{-r}$ rather than $e^{-\theta_b}$, whether the intensity varies across investors or across periods. For a mean number of bidders between 25 and 50, matching the average empty-book probability of Jovanovic and Menkveld (2022) requires r between 3.0 and 4.7, a coefficient of variation of the intensity, $1/\sqrt{r}$, of about one half. The numerical examples use $r = 4$.[24]

[24] Their empty-book probability averages e^{-m} over the days of the sample, so this calibration reflects variation in the intensity across days. Their coefficient of variation of the number of bids, between 0.1 and 0.2, is measured

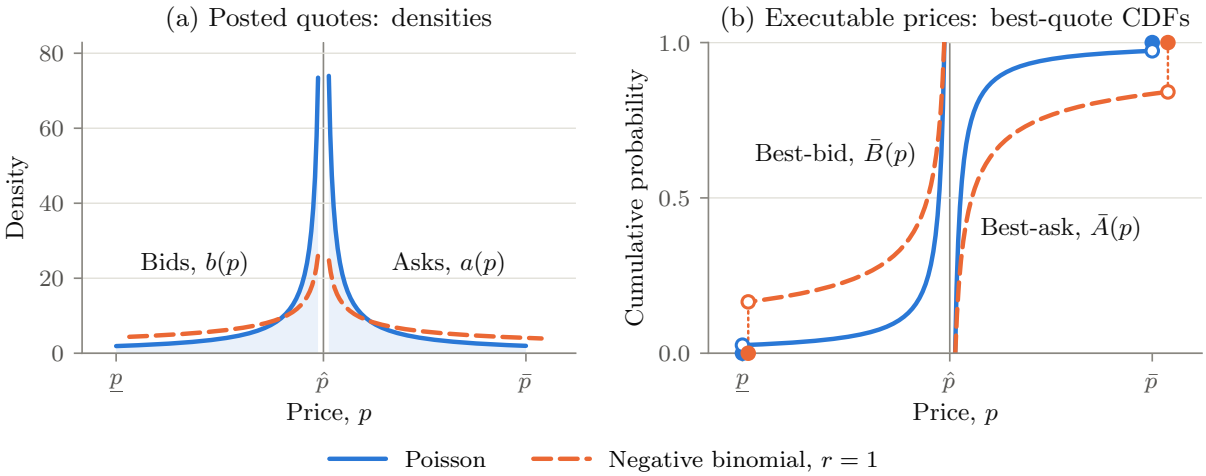


Figure 8: Equilibrium price dispersion with overdispersed quote arrivals

Overdispersion changes the probability that a quote executes, but not the logic of free entry. Free entry (4) and (5) with these execution probabilities therefore yields closed forms that parallel Propositions 1 and 2 and Corollary 2, where $\mathcal{F}$ again denotes the flash crash probability.

Proposition 11. *If $P_k(\theta; r)$ is negative binomial with mean $\theta \in \mathbb{R}_+$ and dispersion $r > 0$, then in the recursive monetary equilibrium:*

(i) Free entry determines the mean numbers of quotes and the flash crash probability,

$$\theta_j = r\left[\left(\frac{\varphi^m \Gamma_j}{\kappa}\right)^{1/(r+1)} - 1\right], \quad j \in \{a, b\}, \quad \text{and} \quad \mathcal{F} = \left(1 + \frac{\theta_b}{r}\right)^{-r} = \left(\frac{\kappa}{\varphi^m \Gamma_b}\right)^{r/(r+1)}.$$

(ii) The ask and bid distributions are

$$A(p) = \frac{r}{\theta_a}\left\{\left[\frac{\varphi^m(p - \hat{p})}{\kappa}\right]^{1/(r+1)} - 1\right\} \quad \text{on } \mathcal{A} \equiv \left[\hat{p} + \frac{\kappa}{\varphi^m}, \bar{p}\right],$$
$$B(p) = 1 - \frac{r}{\theta_b}\left\{\left[\frac{\varphi^m(\hat{p} - p)}{\kappa}\right]^{1/(r+1)} - 1\right\} \quad \text{on } \mathcal{B} \equiv \left[\underline{p}, \hat{p} - \frac{\kappa}{\varphi^m}\right].$$

within a day. Since the squared coefficient of variation of a negative binomial count is $1/\theta + 1/r$, it implies $r \geq 25$ within a day.

(iii) The best-ask and best-bid distributions are

$$\bar{A}(p) = \begin{cases} 1 - \left[\dfrac{\kappa}{\varphi^m (p - \hat{p})}\right]^{r/(r+1)} & \text{if } p \in \left[\hat{p} + \frac{\kappa}{\varphi^m}, \bar{p}\right) \\ 1 & \text{if } p = \bar{p}, \end{cases}$$

$$\bar{B}(p) = \begin{cases} 0 & \text{if } p = \underline{p} \\ \left[\dfrac{\kappa}{\varphi^m (\hat{p} - p)}\right]^{r/(r+1)} & \text{if } p \in \left(\underline{p}, \hat{p} - \frac{\kappa}{\varphi^m}\right], \end{cases}$$

with mass points of size $(1 + \theta_a/r)^{-r}$ *at* $\bar{p}$ *and* $\mathcal{F}$ *at* $\underline{p}$.

(iv) The real (ex-dividend) equity price is

$$\phi^s = \frac{\bar{\beta}_H \eta}{1 - \bar{\beta}_H \eta} \left\{ \varepsilon_H + \frac{\pi_L}{1 - \pi_L} \left[\varphi^m \hat{p} - \frac{\kappa \theta_b}{1 - \mathcal{F}} \right] \right\},$$

and the real value of money φ^m *is the unique solution to*

$$\frac{\varepsilon_H + \phi^s}{1 - (1 + \theta_a/r)^{-r}} \int_{\hat{p} + \frac{\kappa}{\varphi^m}}^{\bar{p}} \frac{r}{r+1} \left[\frac{\kappa}{\varphi^m (p - \hat{p})}\right]^{r/(r+1)} \frac{\mathrm{d}p}{\varphi^m p \, (p - \hat{p})} = \frac{\mu - \bar{\beta}_L}{\bar{\beta}_H}.$$

As $r \to \infty$*, these expressions converge to those of Propositions 1 and 2 and Corollary 2.*

Proposition 11 shows that the Poisson results are a special case. An equilibrium exists and is unique if $\bar{\beta} < \mu < \bar{\mu}_r$, where $\bar{\mu}_r$ equals $\bar{\mu}$ in the Poisson limit and is lower when quotes are overdispersed, and the comparative statics of Propositions 3–6, 7*(i)–(ii)*, 8*(i)* and 9 and of Corollary 3 carry over. What changes is the shape of the distributions. By part *(iii)*, the distance between an executable price and the interbroker price, conditional on trade, follows a truncated Pareto distribution with tail index $r/(r+1)$, against one in the Poisson case. Overdispersion therefore fattens the tail: because quotes cluster, more investors receive few quotes and trade far from the interbroker price.

Figure 8 illustrates, comparing the equilibrium distributions in the Poisson case with those for $r = 1$, a higher degree of overdispersion than the calibrated $r = 4$, chosen so that the difference is easy to see. Overdispersion spreads posted quotes away from the interbroker price, towards the edges of the book, and enlarges the mass points at the edges, the probabilities that

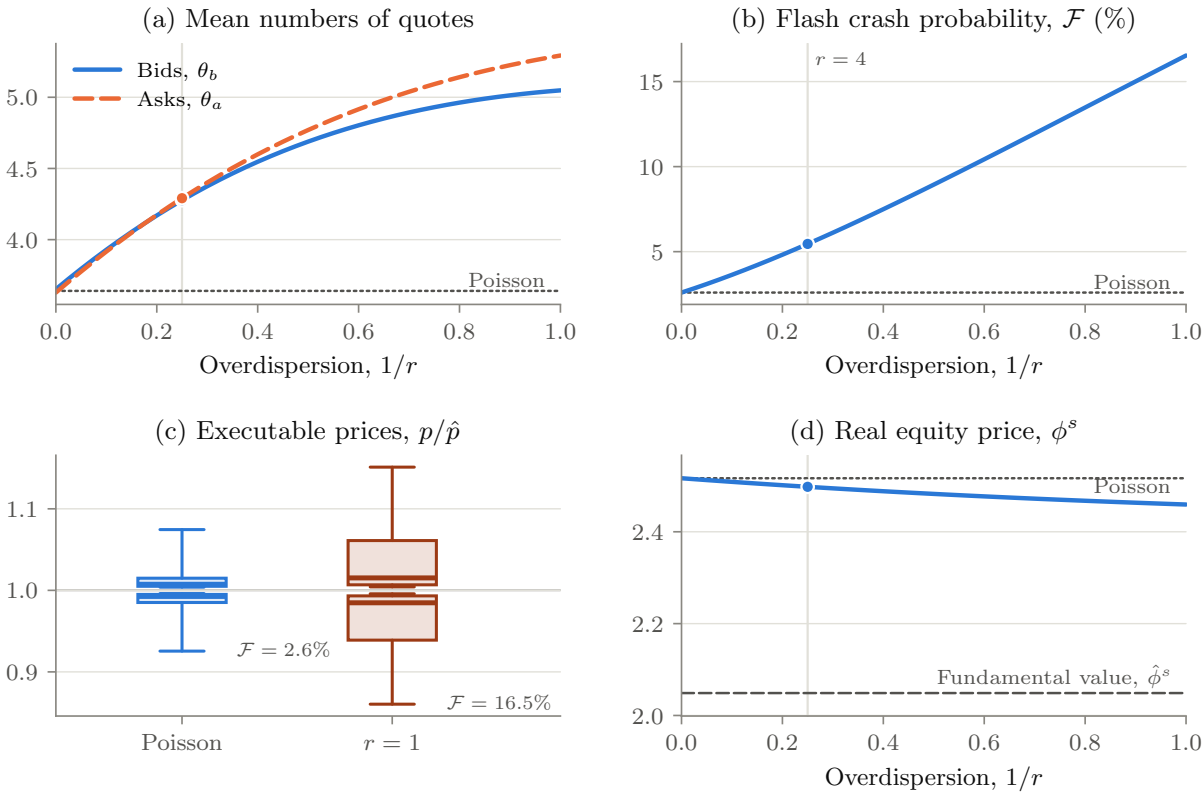


Figure 9: Overdispersed quote arrivals, against the Poisson case. Dotted lines mark the Poisson case and dots $r = 4$; panel (c) compares the Poisson case with $r = 1$.

a seller receives no bid and that a buyer receives no ask.

Figure 9 shows the effects of overdispersion at the baseline policy. One might expect that, because overdispersion raises the mean number of quotes over the range in panel (a), it would make flash crashes less likely. Instead, it makes them more likely (panel (b)). A bid at the bottom of the book is more likely to find a seller with no other bid, so brokers quote more, but quotes cluster on well-connected investors, so more sellers receive none. A market can therefore become more fragile even as quoting rises. Overdispersion also stretches the distribution of executable prices (panel (c)) and lowers the equity price and the speculative premium (panel (d)).

E.2 Overdispersion as a Policy Lever

Market design determines how evenly quotes reach investors, and so acts on r rather than on the posting cost or money growth. Under the made-available-to-trade mandate of the Commod-

ity Futures Trading Commission, for example, a request for quote on a swap execution facility must go to at least three dealers (Riggs et al., 2020); all-to-all platforms and quoting obligations for designated market makers are other examples. Two questions arise. Does overdispersion change optimal monetary policy? And is reducing it useful in its own right? With negative binomial quotes, welfare is given by (21) with $e^{-\theta_b}$ replaced by $\mathcal{F}$.

Corollary 4. *If $P_k(\theta; r)$ is negative binomial with mean $\theta \in \mathbb{R}_+$ and dispersion $r > 0$ and $\varepsilon_H - \varepsilon_L > 2\kappa_0$, then in the recursive monetary equilibrium:*

(i) Welfare is maximised when asks cover only the shares traded and bid intensity is $\theta_b^* = r\{[(\varepsilon_H - \varepsilon_L - \kappa_0)/\kappa_0]^{1/(r+1)} - 1\}$, and equilibrium bid quoting is never excessive, as in Proposition 10(ii).

(ii) The efficient flash crash probability, $\mathcal{F}^* = [\kappa_0/(\varepsilon_H - \varepsilon_L - \kappa_0)]^{r/(r+1)}$, is decreasing with r, and welfare at the efficient quote intensities is increasing with r.

(iii) The Friedman rule is optimal, and the flash crash probability there, $[(\kappa_0 + \tau)/(\varepsilon_H - \varepsilon_L - \kappa_0 - \tau)]^{r/(r+1)}$, equals $\mathcal{F}^*$ if $\tau = 0$, exceeds it if $\tau > 0$, and is decreasing with r.

Corollary 4 answers both questions. First, overdispersion does not change the case for the Friedman rule. The marginal bid lowers the flash crash probability by $(1 + \theta_b/r)^{-(r+1)}$, which is also its probability of executing at the bottom of the book, so overdispersion scales the social and private returns to bidding by the same factor. At the Friedman rule, the real width of the bid side of the book is $\varepsilon_H - \varepsilon_L - \kappa$ whatever the value of r, so bid quoting there is efficient without regulation and insufficient with a regulatory wedge. Second, reducing overdispersion is a lever in its own right. It makes each bid more effective at preventing flash crashes, so the efficient flash crash probability falls and welfare at the efficient quote intensities rises (part *(ii)*), and it lowers the flash crash probability at the Friedman rule (part *(iii)*). Market design can therefore lower flash crash risk even when monetary policy is already at the Friedman rule.

E.3 Optimal Overdispersion

How far should a market designer who maximises welfare reduce overdispersion? The problem has the structure of the long-run model with seller entry in Mangin et al. (2026), whose sellers correspond to bids and whose free-entry condition corresponds to Proposition 11 *(i)*. Entry is efficient in their model, and minimal dispersion maximises surplus. Here, with a regulatory

wedge, the private return to a bid falls short of its social value (Proposition 10*(ii)*), so one might expect a different answer. Since the Friedman rule remains optimal (Corollary 4*(iii)*), this paper studies the choice at the Friedman rule, where ask quoting vanishes and the real width of the bid side, $\varphi^m \Gamma_b = \varepsilon_H - \varepsilon_L - \kappa$, does not depend on r. Let $t \equiv \theta_b / r$, so that the variance-to-mean ratio of the number of bids is $1 + t$, and let $g(t) \equiv (1 + t)\ln(1 + t) - t$, which is positive and strictly increasing for $t > 0$.

Proposition 12. *If $P_k(\theta; r)$ is negative binomial with mean $\theta \in \mathbb{R}_+$ and dispersion $r > 0$ and $\varepsilon_H - \varepsilon_L > 2\kappa_0$, then at the Friedman rule:*

(i) Welfare, $\mathcal{W} = \varepsilon_H - \pi_L(\varepsilon_H - \varepsilon_L)\mathcal{F} - \kappa_0 \pi_L(\theta_b + 1 - \mathcal{F})$, varies with the dispersion parameter according to

$$\frac{d\mathcal{W}}{dr} = \pi_L \left\{ \kappa_0 \, g(t) + (\varepsilon_H - \varepsilon_L) \frac{\tau}{\kappa} \left(-\frac{d\mathcal{F}}{dr} \right) \right\},$$

where $-d\mathcal{F}/dr > 0$, so the first term is positive and the second is positive if $\tau > 0$; welfare is therefore strictly increasing in r and is maximised in the Poisson limit, $r \to \infty$.

(ii) Suppose that market design can raise the dispersion parameter above r_0 at a resource cost of $\xi \geq 0$ per unit of r per period, in units of the aggregate dividend. Welfare net of the cost of design is strictly concave in r and maximised at

$$r^* = \max \left\{ r_0, \ \frac{\ln y}{\ln(1 + t^*)} - 1 \right\},$$

where $y \equiv (\varepsilon_H - \varepsilon_L - \kappa)/\kappa$ and $t^* \geq 0$ solves $\pi_L\{\kappa_0 g(t^*) + [(\varepsilon_H - \varepsilon_L)\tau/(\varepsilon_H - \varepsilon_L - \kappa)](1 + t^*)\ln^2(1 + t^*)/\ln y\} = \xi$. Without a cost of design, $t^* = 0$ and welfare is maximised in the Poisson limit.

Proposition 12*(i)* shows that, at the Friedman rule and without a cost of design, the Poisson limit maximises welfare, as in Mangin et al. (2026). Part *(i)* separates two effects of more even quote arrivals. The first, $\kappa_0 g(t)$, is always positive. With overdispersion, quotes pile up on well-connected investors, where an additional quote lowers the price but creates no trade, while other investors receive none; spreading quotes more evenly lets more investors trade for a given number of quotes. Without regulation, bid entry at the Friedman rule is efficient, and this is the only effect, exactly as in Mangin et al. (2026), where entry is also efficient. The second

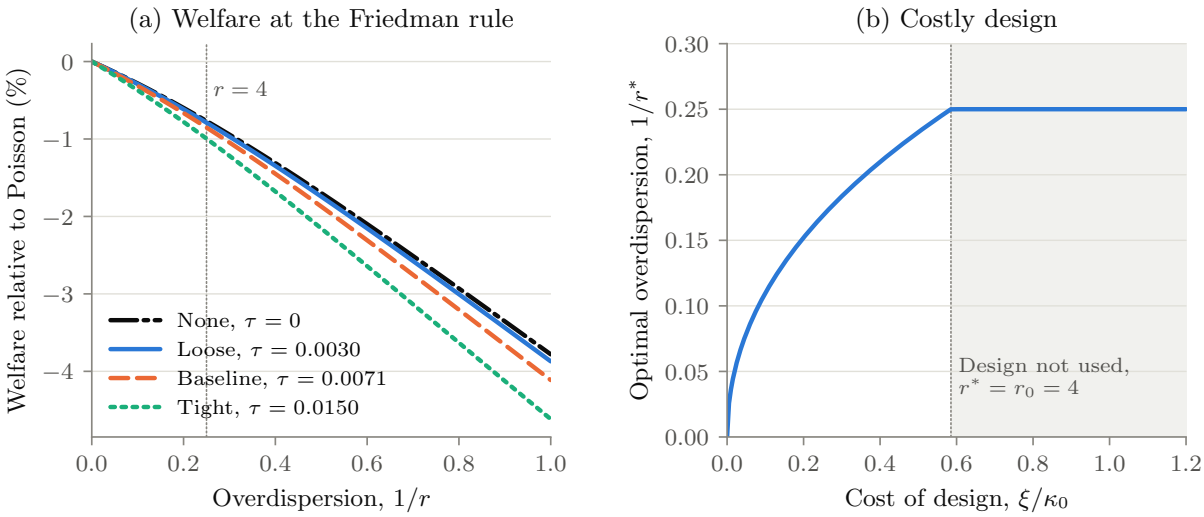


Figure 10: Welfare-optimal overdispersion at the Friedman rule. Panel (b) uses the baseline regulatory wedge.

effect is proportional to the regulatory wedge. A wedge makes bid quoting insufficient (Proposition 10*(ii)*), so the additional trade that more even arrivals create is worth more to society than to brokers, and the wedge strengthens the case for even arrivals. Panel (a) of Figure 10 illustrates: in the numerical example, welfare at the Friedman rule falls with overdispersion without regulation and under each regulatory wedge of Figure 7.

In practice, market design is costly. Part *(ii)* adds a resource cost of design, such as the cost of running a trading platform or enforcing a quoting mandate. The designer raises r until the marginal benefit, which falls as arrivals become more even, equals the marginal cost ξ. Given the posting cost, the cost of design pins down the variance-to-mean ratio of bid arrivals, $1 + t^*$, which rises with ξ, and the real width of the bid side at the Friedman rule then determines the r^* that delivers it. The optimal dispersion parameter therefore falls with the cost of design and grows without bound, at the rate $\xi^{-1/2}$, as that cost vanishes. In the numerical example, it also rises with the regulatory wedge: a larger wedge makes bid quoting more insufficient, which makes more even arrivals more valuable. Panel (b) of Figure 10 illustrates: market design is used only when its cost is low enough, and below that threshold optimal overdispersion rises with the cost of design, reaching the Poisson limit only when design is free.